北京交通大学

# 本科毕业论文（设计）

## Mamyshev 振荡器的性能影响因素研究

## Performance-Determining Factors in Mamyshev Oscillators


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# 中文摘要

**摘要：** Mamyshev 振荡器是一种新型光纤激光器，其利用光纤中自相位调制引起的光谱展宽，并结合偏移光学滤波器对高强度脉冲进行筛选，从而产生高能量超短脉冲。与基于非线性偏振旋转（Nonlinear Polarization Rotation, NPR）或真实可饱和吸收体（Saturable Absorber, SA）的锁模技术相比，Mamyshev 机制具有更高的调制深度、更强的环境稳定性，并且兼容全光纤集成，因此被认为是下一代超快光纤激光器的重要候选方案。本文通过理论建模与数值仿真的方法，研究了环形腔和线性腔 Mamyshev 振荡器的性能影响因素。腔内脉冲演化过程采用广义非线性薛定谔方程（Generalized Nonlinear Schrödinger Equation, GNLSE）进行描述，并综合考虑色散、非线性效应、增益动力学以及光谱滤波等因素。在数值求解部分，本文采用分步傅里叶方法（Split-Step Fourier Method, SSFM）和四阶 Runge-Kutta 相互作用绘景法构建了环形腔 Mamyshev 结构的仿真框架。基于该模型，本文分析了滤波器中心波长间隔、滤波器带宽、掺镱光纤长度以及泵浦功率对脉冲输出特性的影响。本研究可为基于 Mamyshev 结构的超快光纤激光器设计提供理论依据和实际参考。

**关键词：** Mamyshev 振荡器；锁模；超快光纤激光器；非线性脉冲传输；广义非线性薛定谔方程；分步傅里叶方法；光谱滤波

# ABSTRACT

**ABSTRACT:**

The Mamyshev oscillator is a new type of fiber laser, which utilizes the spectral broadening caused by self-phase modulation in the fiber and offset optical filters to select high-intensity pulses, thereby generating high-energy ultrashort pulses. Compared with mode-locking techniques based on Nonlinear Polarization Rotation (NPR) or real Saturable Absorbers (SA), the Mamyshev mechanism possesses higher modulation depth, stronger environmental robustness, and compatibility with all-fiber integration, making it an ideal platform for the next generation of ultrafast fiber lasers. In this paper, the performance-determining factors of ring-cavity and linear-cavity Mamyshev oscillator are studied through theoretical modeling and numerical simulation. The pulse evolution inside the cavity is modeled by using the Generalized Nonlinear Schrödinger Equation (GNLSE), considering dispersion, nonlinearity, gain dynamics, and filtering effects. In the numerical solution part, the Split-Step Fourier Method (SSFM) and the fourth-order Runge-Kutta method are adopted to construct the simulation framework of the ring-cavity Mamyshev structure. The influence of filter center-wavelength spacing, filter bandwidth, Yb-doped fiber length, and pump power on the pulse output are analyzed. This study provides a theoretical basis and practical guidance for the design of ultrafast fiber lasers based on the Mamyshev structure.

**KEYWORDS:** Mamyshev oscillator; mode-locking; ultrafast fiber lasers; nonlinear pulse propagation; GNLSE; SSFM; spectral filtering

# 目　　录

**中文摘要** ..... I
ABSTRACT ..... II
**目　　录** ..... III
1 Introduction ..... 1
2 Background ..... 3
2.1 Passive Mode-Locking Techniques ..... 3
2.1.1 Mode-Locking with Real Saturable Absorbers ..... 3
2.1.2 Mode-Locking with Equivalent Saturable Absorbers ..... 4
3 Principles of Numerical Modeling ..... 11
3.1 Pulse Propagation in Optical Fibers ..... 11
3.2 Dispersion in Optical Fibers ..... 13
3.3 Fiber Gain ..... 13
3.4 Nonlinear Effects ..... 14
3.5 Spectral Filtering in Mamyshev Oscillators ..... 14
3.6 Split-Step Fourier Method (SSFM) ..... 15
3.7 Runge–Kutta in the Interaction Picture (RK4IP) ..... 16
4 Simulation and Analysis of Mamyshev Oscillator Structures ..... 18
4.1 Simulation and Analysis of the Ring-Cavity Mamyshev Oscillator ..... 18
4.1.1 Pulse Evolution and Output Characteristics in the Oscillator ..... 19
4.1.2 Influence of Pump Power on Pulse Characteristics ..... 21
4.1.3 Influence of filter offset on Pulse Characteristics ..... 23
4.1.4 Influence of filter bandwidth on Pulse Characteristics ..... 25
4.1.5 Influence of Gain Fiber Length on Pulse Characteristics ..... 28
4.2 Simulation and Analysis of linear-cavity Mamyshev Oscillator ..... 30
4.2.1 Pulse Evolution and Output Characteristics in the Oscillator ..... 30
4.2.2 Influence of Pump Power on Pulse Characteristics ..... 32
4.2.3 Influence of filter offset on Pulse Characteristics ..... 34
4.2.4 Influence of filter bandwidth on Pulse Characteristics ..... 36

4.2.5 Influence of Gain Fiber Length on Pulse Characteristics ..... 37
5 Conclusion ..... 39
**参考文献** ..... 42
**致　　谢** ..... 44

# 1 Introduction

Since 1960 when the first ruby laser was successfully developed, the laser has rapidly evolved with its excellent monochromaticity, directionality, and high brightness characteristics, becoming an important cornerstone of modern science and industry[1]. Entering the 1980s, with the progress of mode-locking technology, the time scale of laser pulses was continuously compressed, entering from the picosecond ($10^{-12}$ s) to the femtosecond ($10^{-15}$ s) era.

Mode-locking is achieved by locking the phases of multiple longitudinal modes inside the laser cavity to make them coherently superimposed, thus forming high-peak ultrashort pulse output in the time domain. Based on different implementation methods, mode-locking can be divided into active mode-locking and passive mode-locking. The former relies on active devices such as electro-optic or acousto-optic modulators to periodically modulate the intracavity light field, while the latter utilizes passive devices such as Saturable Absorbers (SA) to realize spontaneous mode-locking through the difference in transmittance dependent on light intensity. In comparison, the passive mode-locking structure is more concise, lower in cost, and higher in stability, so it has become the mainstream implementation method for ultrashort pulse lasers[2].

However, traditional passive mode-locking techniques still face several limitations. Physical Saturable Absorbers (SAs) are prone to optical damage and suffer from limited modulation depth under high-power operation. While equivalent saturable absorber, such as Nonlinear Polarization Rotation (NPR) and Nonlinear Optical Loop Mirrors (NOLM), can circumvent material damage, their performance is typically sensitive to environmental fluctuations. These constraints have limited the improvement of pulse energy and stability in ultrafast fiber lasers. Mamyshev oscillators offer a highly promising solution to these issues. Based on Self-Phase Modulation (SPM) and the use of offset filters, this architecture functions as an artificial saturable absorber with a high modulation depth. Since low-intensity noise cannot broaden its spectrum sufficiently to pass through the offset filters, only high-intensity pulses are selectively transmitted. Consequently, the Mamyshev mechanism supports the generation of high-energy and ultrashort pulses. Additionally, its compatibility with all-fiber configurations provides a distinct advantage for engineering applications. From an experimental and engineering standpoint, the output performance of a Mamyshev oscillator is strongly influenced by various cavity parameters, including pump power, filter bandwidth, spectral offset, and gain fiber length. Direct experimental optimization of these parameters can be time-consuming and labor-intensive, as each variable affects the intricate balance between gain, dispersion, nonlinear phase accu-

mulation, and spectral filtering. Therefore, numerical simulation provides an effective method for understanding intracavity pulse evolution and predicting suitable parameter ranges before experimental implementation.

In this paper, we numerically investigate both ring and linear Mamyshev oscillator configurations. Intracavity pulse propagation is described by the Generalized Nonlinear Schrödinger Equation (GNLSE), solved using the split-step Fourier method and a fourth-order Runge-Kutta algorithm. With pump power, filter center wavelength spacing, filter bandwidth, and Ytterbium-doped gain fiber length as the fundamental parameters, we analyze how their variations affect pulse energy, peak power, pulse width, and output wavelength. This study aims to clarify the performance determinants of Mamyshev oscillators and provide theoretical guidance for the design and optimization of high-energy ultrafast fiber lasers.

# 2 Background

## 2.1 Passive Mode-Locking Techniques

The core mechanism of passive mode-locking is the saturable absorption effect. When an ultrashort pulse passes through a saturable absorber, the SA will quickly absorb the high-intensity pulse and reach saturation, leading to a significant increase in its transmittance. On the contrary, the low-intensity components cannot make it reach saturation absorption, and the transmittance is greatly weakened. After multiple round trips inside the laser cavity, this intensity-dependent transmittance will cause the low-intensity pulses to gradually decay, while the high-intensity pulses will survive in the cavity. Finally, this process will form stable, high-peak-power ultrashort pulses.

Passive mode-locking technology can be roughly divided into real saturable absorber mode-locking and equivalent saturable absorber mode-locking. Real saturable absorbers rely on physical materials, such as the inherent saturable absorption properties of Semiconductor Saturable Absorber Mirrors (SESAM) or two-dimensional materials. The equivalent fast saturable absorber utilizes nonlinear optical effects inside the cavity to generate an intensity-dependent transmission function, thereby simulating the behavior of a real saturable absorber. Compared with active mode-locking, these mechanisms reduce the influence of environmental factors and the instability of artificial modulation.

### 2.1.1 Mode-Locking with Real Saturable Absorbers

The development of real saturable absorber mode-locking technology can be traced back to 1966, when DeMaria *et al.* first demonstrated using organic dyes as saturable absorbers to realize passive mode-locking on $Nd^{3+}$-doped glass laser[3]. Although this work marked the birth of passive mode-locking technology, the resulting pulse train exhibited amplitude modulation and operated in a regime later recognized as Q-switched mode-locking rather than true steady-state ultrafast operation. In 1996, Keller et al. developed the SESAM technology from the amplitude resonant coupling cavity, which thoroughly solved the Q-switched mode-locking problem in solid-state lasers[4]. In 2009, Zhang *et al.* reported the first case of ultrafast laser mode-locking realized by using graphene, opening a new era of integrated application of two-dimensional materials and ultrafast lasers[5].

The nonlinear absorption mechanism is when the incident light intensity is lower, electrons in the valence band absorb photons and undergo interband transition to the conduction band, and subsequently relax back to the dirac point and recombine. Under high light intensity, the number of carriers in the conduction band increases significantly, leading to a decrease in the number of electrons available for excitation in the valence band, thereby reducing the absorption efficiency. This process is the saturable absorption mechanism. In recent years, many saturable absorber materials with great potential have emerged, including perovskites[6], antimonene[7], and bismuthene[8]. These materials exhibit various unique electronic structures and nonlinear optical responses, indicating that there are still many candidate materials with excellent saturable absorption characteristics waiting to be discovered.

### 2.1.2 Mode-Locking with Equivalent Saturable Absorbers

The equivalent saturable absorbers utilizes the intracavity nonlinear effects to simulate the absorption mechanism of the real saturable absorber. Different from the real saturable absorber, these mechanisms rely entirely on the nonlinear effects of the fiber, such as Self-Phase Modulation (SPM), Cross-Phase Modulation (XPM), multi-path interference, or spectral filtering to realize effective pulse intensity selection. This section will discuss three representative technologies: NPR, NOLM, and the Mamyshev mechanism.

**Nonlinear Polarization Rotation (NPR).** NPR is one of the earliest and most influential equivalent saturable absorber mechanisms for fiber lasers. The working principle of the equivalent saturable absorber based on NPR relies on the rotation of the State of Polarization (SOP) caused by the Kerr effect inside the fiber. When a pulse propagates in a single-mode fiber, two orthogonal polarization components will experience different nonlinear phase shifts, which is caused by SPM and XPM. This intensity-dependent phase imbalance will cause the SOP to rotate when the pulse propagates along the fiber.

As shown in Fig. 2-1, the linearly polarized pulse emitted from the polarizer at point A is first converted into an elliptically polarized state by PC1. When the pulse propagates in the single-mode fiber, the Kerr effect will induce the rotation of its SOP. The part with higher intensity experiences a larger nonlinear phase shift, therefore undergoing stronger polarization rotation. After passing through PC2, the pulse reaches the output polarizer. Only the components with polarization direction aligned with the transmission axis can pass through, while the remaining components are filtered out, thereby generating effective saturable absorption

characteristics.

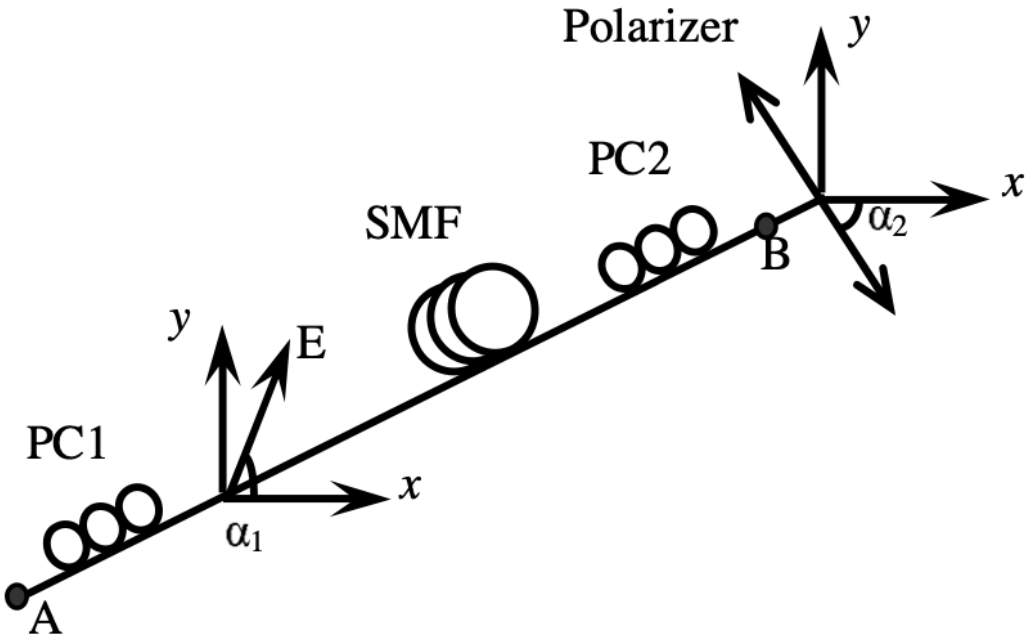



Figure 2-1 Operation principle of NPR[9]

In 1992, Matsas *et al.* first reported the passive mode-locking of an all-fiber ring laser based on NPR[10]. They utilized a pump power of 70 mW to realize self-starting mode-locking and generated soliton pulses of 1.6 ps, thereby establishing NPR as a practical and robust Equivalent Saturable Absorber. Subsequent research optimized the waveplate structure, dispersion management, and cavity birefringence, extending the working range of NPR from anomalous dispersion to normal dispersion, and realized the generation of dissipative solitons.

**Nonlinear Optical Loop Mirror (NOLM).** The NOLM utilizes the intensity-dependent interference caused by the Kerr effect inside the Sagnac fiber loop to act as an equivalent saturable absorber. When a pulse propagates in the nonlinear fiber loop, the Kerr effect will induce an intensity-dependent refractive index change, leading to different nonlinear phase differences generated by the counter-propagating components. When recombining at the coupler, this phase difference determines whether the interference is constructive or destructive, thereby resulting in the transmittance increasing with the increase of pulse intensity.

As illustrated in Fig. 2-2, the input pulse is split into clockwise and counter-clockwise components by the coupler. The pulse propagating in the nonlinear fiber will accumulate a phase difference caused by the Kerr effect proportional to its instantaneous intensity, while the

reference pulse accumulates a smaller phase difference. When these two components recombine at the coupler, constructive interference enhances the high-intensity components, while destructive interference suppresses the low-intensity components. This intensity-dependent transmission simulates saturable absorption and can realize stable passive mode-locking in the laser cavity.

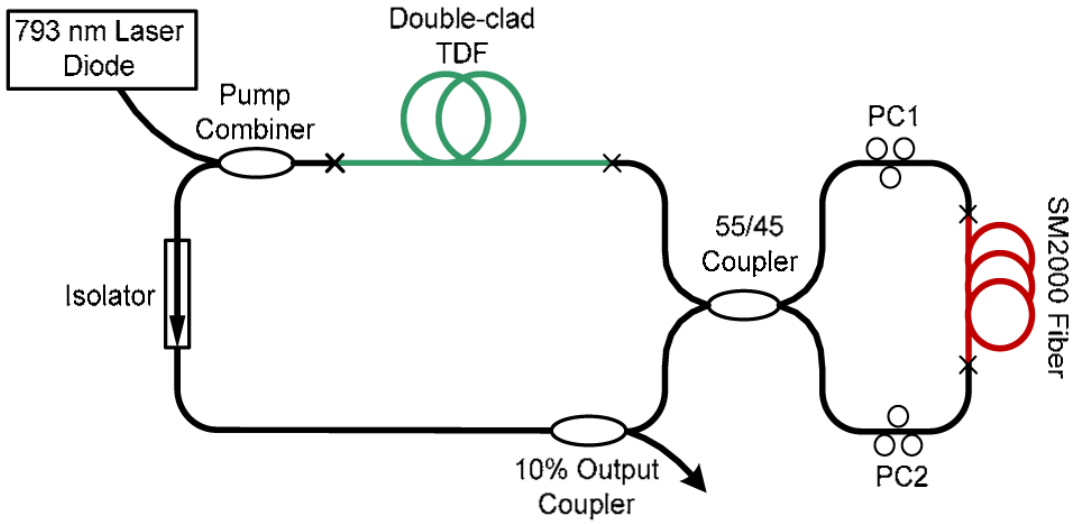



Figure 2-2 Schematic of the "figure-of-eight" mode-locked double-clad TDF laser based on NOLM[11].

**Mamyshev Mode-Locking** Mamyshev mode-locking provides an equivalent saturable absorber based on the combination of strong SPM and offset spectral filtering. A Mamyshev oscillator consists of two Mamyshev regenerators arranged in a loop, each composed of a gain fiber, an output coupler, and a bandpass filter. The key requirement is that the two filters exhibit different central wavelengths. As illustrated in Fig. 2-3, two pulses $P_1$ and $P_2$ with the same input center wavelength $\lambda_0$ but different peak powers undergo different degrees of SPM-induced spectral broadening. The stronger pulse $P_1$ broadens sufficiently to extend into the passband of the offset filter at $\lambda_0 + \Delta\lambda$, whereas the weaker pulse $P_2$ does not. After filtering, $P_1$ is transmitted while $P_2$ is rejected, yielding an intensity-dependent transmission equivalent to saturable absorption.

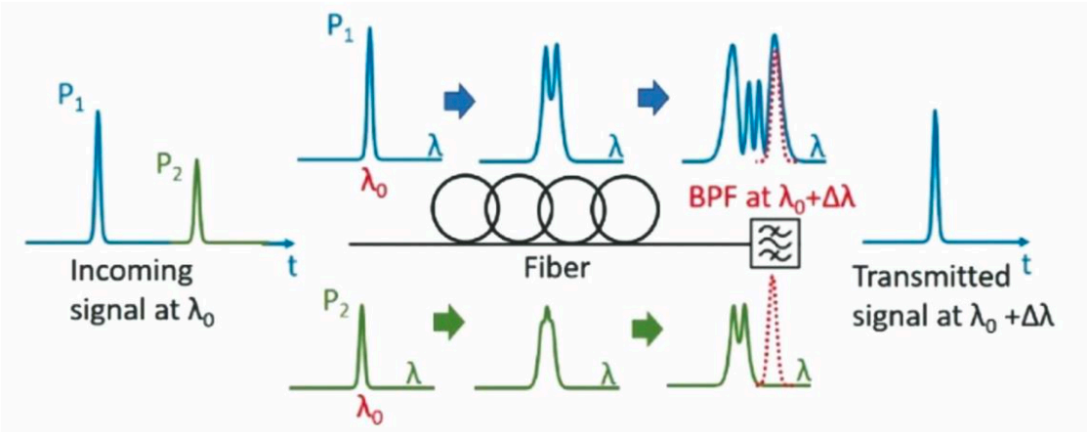



Figure 2-3 Principle of Mamyshev mode-locking based on SPM broadening and offset spectral filtering[12].

Although the concept of mode-locking using offset spectral filtering was first proposed in 1994[12], it attracted limited attention at the time. Interest in this mechanism revived in recent years. In 2017, Liu *et al.* demonstrated an environmentally stable ultrafast source based on cascaded Mamyshev regeneration, as illustrated in Fig. 2-4. In this design, two sequential filter–gain segments are employed to enhance spectral broadening. Simulation results indicated that a Mamyshev oscillator seeded by a low-energy ($<$ 10 ps) pulse could generate mode-locked pulses with energies up to 190 nJ and dechirped durations below 20 fs. Experimentally, the system produced $\sim$ 50 nJ pulses with $\sim$ 40 fs duration at a repetition rate of 17 MHz, showing good qualitative agreement with the theoretical predictions[13].

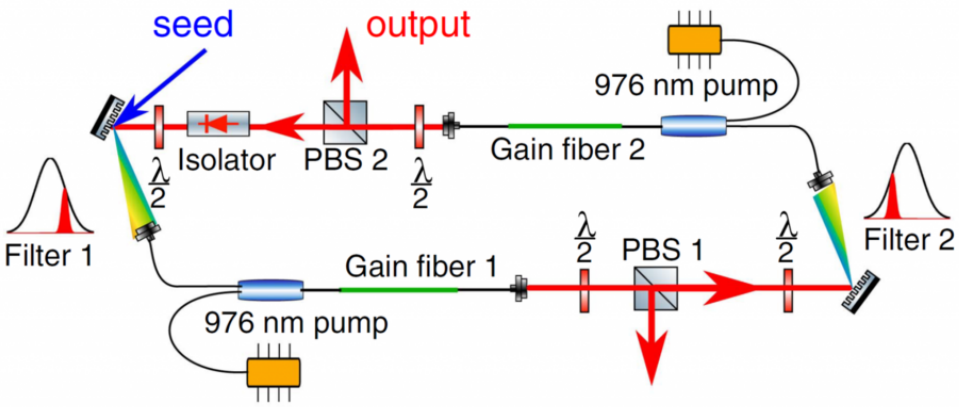



Figure 2-4 Schematic of the cascaded Mamyshev oscillator based on two offset spectral filtering stages. The system contains two successive Mamyshev regeneration stages, where gain-induced amplification, self-phase-modulation-based spectral broadening, and offset spectral filtering occur sequentially.[13].

Further energy scaling was achieved with cascaded Mamyshev stages, allowing stronger spectral broadening and deeper modulation. Using this approach, pulse energies reached 190 nJ with 35 fs duration. However, these oscillators still lacked reliable self-starting. Sidorenko *et al.* introduced a startup arm using a flippable mirror to provide a transient high-power perturbation during the noise build-up stage, enabling pulses to cross the offset filters and initiate mode-locking with greatly improved repeatability[14]. The experimental setup is shown in Fig. 2-5. This design achieved multi-megawatt peak power while maintaining excellent operational stability.

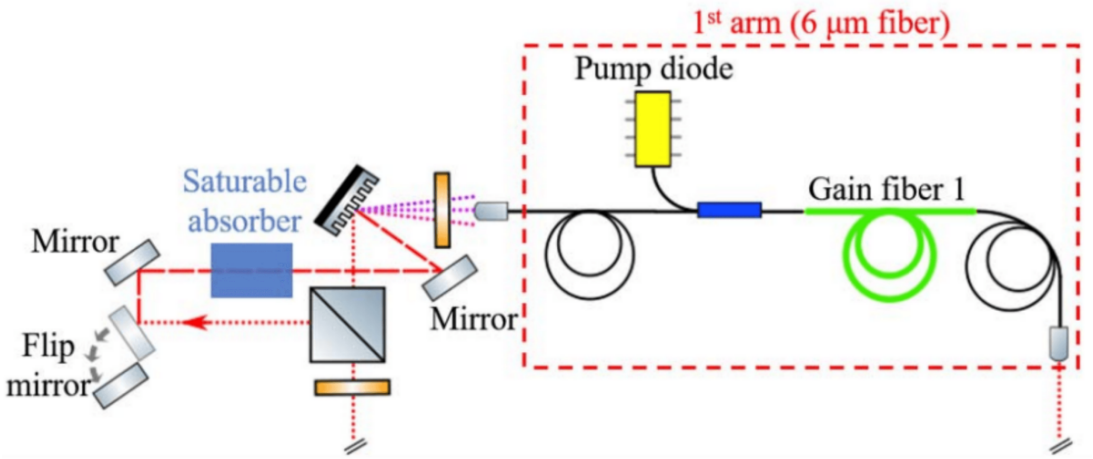



Figure 2-5 Experimental setup of the self-seeded Mamyshev oscillator with a startup arm. A flippable mirror is used to temporarily direct the intracavity pulse into an auxiliary arm containing a saturable absorber, which provides a high-intensity perturbation during the initial noise build-up stage.[14]

As research shifted toward pushing the energy limit further, excessive nonlinear phase accumulation became a major challenge. Wu *et al.* addressed this by incorporating Large-Mode-Area Photonic Crystal Fiber (LMA-PCF), thereby reducing nonlinear phase per unit energy. Their Mamyshev oscillator reached 1.1 $\mu$J pulse energy, 41 fs duration, and 13 MW peak power[15], marking an important milestone in high-energy ultrafast fiber laser development.

Beyond energy scaling, bandwidth expansion became another frontier. Ma *et al.* demonstrated a Mamyshev oscillator employing highly nonlinear photonic crystal fiber and dispersion-delay lines, enabling self-similar nonlinear evolution combined with offset filtering. The system produced a 394 nm (20 dB) spectral bandwidth and 17 fs compressed pulses, achieving few-cycle operation[16]. This result revealed the potential of the Mamyshev architecture for generating ultrabroadband ultrashort pulses.

In 2019, As shown in Fig. 2-6, Olivier *et al.* developed a femtosecond Mamyshev fiber os-

cillator operating at 1550 nm, achieving a maximum single-pulse energy of 31.3 nJ and a pulse width of approximately 100 fs after a grating compressor[17]. This research demonstrates that the Mamyshev oscillator is not limited to 1 $\mu$m Yb-doped fiber systems; its SPM mechanism can also be applied to Er-doped fiber systems. Since 1550 nm is a commonly used wavelength in optical communication, the significance of this work lies not only in expanding the operating wavelength range of the Mamyshev oscillator but also in illustrating the potential application value of this mechanism in communication-band ultrashort pulse sources, nonlinear optical experiments, and subsequent fiber amplification systems.

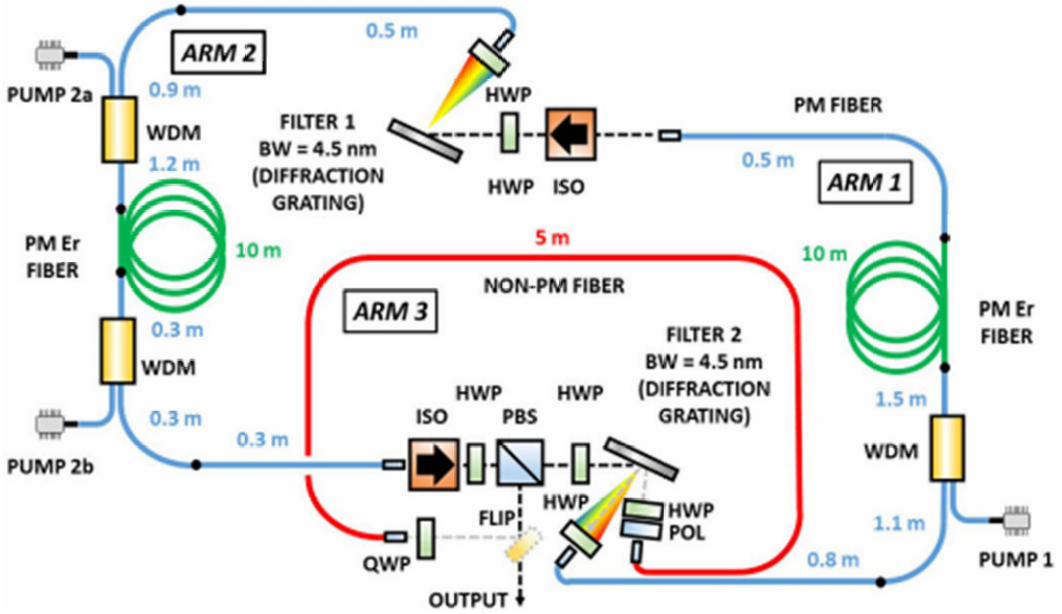



Figure 2-6 Experimental configuration of the 1550 nm Er-doped Mamyshev oscillator. The two offset diffraction-grating filters and Er-doped gain sections provide Mamyshev regeneration through spectral broadening and intensity-dependent filtering, showing the applicability of this mechanism in the communication wavelength band.[17]

As the energy of femtosecond pulses generated by Mamyshev oscillators gradually increases, subsequent research has shifted its focus from simply improving output performance to enhancing system stability and startup reliability. Traditional Mamyshev oscillators rely on two spectral filters with separated center wavelengths to achieve equivalent saturable absorption. While this structure effectively suppresses low-intensity noise and continuous wave components, it also introduces startup difficulties. Because low-intensity noise does not initially generate sufficient SPM spectral broadening to bridge the spectral gap between the two filters, the system often cannot directly self-start from the noise and requires additional startup mechanisms or dynamic adjustment processes. To address this issue, Chen *et al.* studied the

startup dynamics of linear cavity femtosecond Mamyshev oscillators[18]. This work employed pump modulation and movable spectral filters to assist startup. Experimental results show that by reducing the filter gap and adjusting the pump modulation conditions during startup, and then restoring to the target filtering state after pulse formation, the linear cavity Mamyshev oscillator can enter stable mode-locked operation. The system ultimately outputs a 21 nJ pulse, and after external compression, it achieves a pulse width of approximately 65 fs.

In 2023, Piché's group realized an all-polarization-maintaining as shown in Fig. 2-7, linear-cavity Mamyshev oscillator using Fiber Bragg Gratings (HR-FBG and LR-FBG) as end mirrors[19]. Their system self-started under single-pump excitation and delivered 102 nJ, 37 fs compressed pulses at 52 MHz repetition rate, corresponding to peak power exceeding 2 MW. Remarkably, the oscillator demonstrated 100% self-starting success across more than one hundred trials and maintained stable single-pulse operation across a wide pump range, marking a significant step toward fully practical, highly robust Mamyshev-based ultrafast lasers.

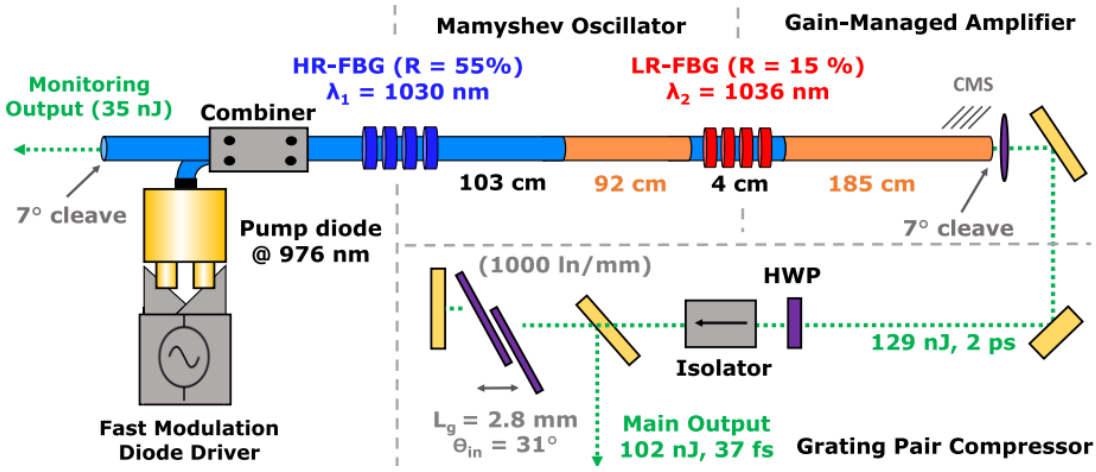



Figure 2-7 Linear cavity Mamyshev laser setup.[19]

# 3 Principles of Numerical Modeling

This chapter presents the theoretical foundations of numerical modeling for Mamyshev oscillators. To understand how a mode-locked pulse evolves within an optical resonator, it is essential to examine the dominant factors governing its spatiotemporal dynamics, including fiber gain, group-velocity dispersion, and Kerr nonlinearity. Whether the pulse can reach a stable mode-locked state depends on whether these competing mechanisms can reach a balance over many cavity round trips.

Before analyzing the steady-state behavior and performance of Mamyshev oscillators, we therefore establish a pulse propagation equation that incorporates gain, dispersion, and nonlinear effects. A rigorous solution of this equation provides the theoretical basis for understanding the intracavity pulse dynamics and for evaluating the conditions required for stable Mamyshev mode-locking.

## 3.1 Pulse Propagation in Optical Fibers

As in all electromagnetic phenomena, light propagation inside optical fibers is governed by Maxwell's equations. In a nonmagnetic, isotropic dielectric medium, these equations can be written as

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{3-1}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \tag{3-2}$$

$$\nabla \cdot \mathbf{D} = \rho_f, \tag{3-3}$$

$$\nabla \cdot \mathbf{B} = 0. \tag{3-4}$$

Here, $\mathbf{E}$ and $\mathbf{H}$ denote the electric and magnetic field vectors, and $\mathbf{D}$ and $\mathbf{B}$ represent the electric and magnetic flux densities, respectively. $\mathbf{J}$ is the free current density and $\rho_f$ is the free charge density. In silica fibers, there are no free charges or currents:

$$\mathbf{J} = 0, \qquad \rho_f = 0.$$

The constitutive relations linking the fields and flux densities are

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \tag{3-5}$$

$$\mathbf{B} = \mu_0 \mathbf{H} + \mathbf{M}, \tag{3-6}$$

where $\varepsilon_0$ and $\mu_0$ are the vacuum permittivity and permeability, and $\mathbf{P}$ and $\mathbf{M}$ denote the electric and magnetic polarization densities. For a nonmagnetic fiber medium,

$$\mathbf{M} = 0, \qquad \mathbf{B} = \mu_0 \mathbf{H}.$$

Substituting Eqs. (3-2)–(3-6) into Eq. (3-1) and using the vector identity

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}, \tag{3-7}$$

the electric field satisfies the vector wave equation:

$$\nabla \times \nabla \times \mathbf{E} = -\frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2}, \tag{3-8}$$

where $c = 1/\sqrt{\varepsilon_0 \mu_0}$ is the speed of light in vacuum. Equation (3-8) shows that both the linear and nonlinear components of the polarization $\mathbf{P}$ determine the essential features of pulse propagation in optical fibers.

To describe ultrashort pulse evolution, the electric field is expressed in the slowly varying envelope form:

$$\mathbf{E}(x,y,z,t) = \frac{1}{2}\left[A(z,t)\,\mathbf{F}(x,y)\,e^{i(\beta_0 z - \omega_0 t)} + \text{c.c.}\right], \tag{3-9}$$

where $A(z,t)$ is the slowly varying complex envelope and $\mathbf{F}(x,y)$ is the transverse mode profile ($\text{LP}_{01}$ mode for single-mode fibers).

The slowly varying envelope approximation (SVEA) assumes

$$\left|\frac{\partial^2 A}{\partial z^2}\right| \ll \beta_0 \left|\frac{\partial A}{\partial z}\right|, \tag{3-10}$$

allowing the removal of second-order derivatives in $z$ and reducing the full vector wave equation to a scalar envelope propagation equation. After projection onto the fiber mode, the general form in the frequency domain becomes

$$\frac{\partial \tilde{A}(z,\omega)}{\partial z} = i\beta(\omega)\tilde{A}(z,\omega) + \tilde{N}(z,\omega), \tag{3-11}$$

where $\tilde{A}(z,\omega)$ is the Fourier transform of $A(z,t)$ and $\tilde{N}(z,\omega)$ represents the contribution from material polarization, including gain and nonlinear effects.

Equation (3-11) serves as the starting point for constructing the nonlinear Schrödinger equation (NLSE). The next sections incorporate dispersion (from $\beta(\omega)$), gain, and Kerr nonlinearity into this propagation framework.

## 3.2 Dispersion in Optical Fibers

Fiber dispersion arises from the frequency dependence of the propagation constant $\beta(\omega)$. Expanding $\beta(\omega)$ near the carrier frequency $\omega_0$ yields

$$\beta(\omega) = \beta_0 + \beta_1(\omega-\omega_0) + \frac{\beta_2}{2}(\omega-\omega_0)^2 + \frac{\beta_3}{6}(\omega-\omega_0)^3 + \cdots, \tag{3-12}$$

where

$$\beta_k = \left.\frac{d^k\beta}{d\omega^k}\right|_{\omega_0}.$$

Here, $\beta_1$ corresponds to group delay, $\beta_2$ is the group-velocity dispersion (GVD), and $\beta_3$ denotes third-order dispersion (TOD).

Introducing the retarded time frame

$$T = t - \beta_1 z \tag{3-13}$$

eliminates first-order dispersion. In the time domain, the linear dispersive contribution becomes

$$\frac{\partial A}{\partial z} = i\frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2} + \frac{\beta_3}{6}\frac{\partial^3 A}{\partial T^3} + \cdots - \frac{\alpha}{2}A, \tag{3-14}$$

where $\alpha$ is fiber loss. Equation (3-14) forms the linear part of the NLSE.

## 3.3 Fiber Gain

In rare-earth–doped fibers, stimulated emission introduces optical gain. The gain coefficient is typically modeled as a saturable function of pulse energy:

$$g(z) = \frac{g_0}{1+E/E_{\text{sat}}}, \qquad E(z) = \int_{-\infty}^{+\infty} |A(z,T)|^2 dT, \tag{3-15}$$

where $g_0$ is the small-signal gain and $E_{\text{sat}}$ is the saturation energy.

The gain contribution to the propagation equation is thus

$$\left.\frac{\partial A}{\partial z}\right|_{\text{gain}} = \frac{g(z)}{2}A. \tag{3-16}$$

Including dispersion and gain, the evolution equation becomes

$$\frac{\partial A}{\partial z} = \frac{g(z)-\alpha}{2}A + i\sum_{k\geq 2}\frac{\beta_k}{k!}\left(i\frac{\partial}{\partial T}\right)^k A + N_{\text{NL}}(z,T), \tag{3-17}$$

where $N_{\text{NL}}$ denotes nonlinear effects yet to be added.

## 3.4 Nonlinear Effects

The dominant nonlinear contribution in silica fibers arises from the third-order polarization

$$\mathbf{P}^{(3)}(t) = \varepsilon_0 \chi^{(3)} |\mathbf{E}(t)|^2 \mathbf{E}(t), \tag{3-18}$$

which produces an intensity-dependent refractive index:

$$n = n_0 + n_2 I, \qquad I \propto |A|^2. \tag{3-19}$$

Under SVEA, this nonlinear polarization yields the Kerr term in the envelope equation:

$$\left.\frac{\partial A}{\partial z}\right|_{\mathrm{NL}} = i\gamma |A|^2 A, \qquad \gamma = \frac{n_2 \omega_0}{c A_{\mathrm{eff}}}, \tag{3-20}$$

which corresponds to SPM.

Combining dispersion, gain, loss, and Kerr nonlinearity leads to the generalized nonlinear Schrödinger equation (GNLSE):

$$\frac{\partial A(z,T)}{\partial z} = \frac{g(z)-\alpha}{2} A + i \sum_{k\geq 2} \frac{\beta_k}{k!} \left( i \frac{\partial}{\partial T} \right)^k A + i\gamma |A|^2 A. \tag{3-21}$$

Equation (3-21) provides the complete model for pulse propagation in the fiber segments of Mamyshev oscillators. Combined with spectral filtering and cavity boundary conditions, it forms the basis for numerical simulations using the split-step Fourier method (SSFM) and the fourth-order Runge–Kutta interaction-picture (RK4IP) algorithm.

## 3.5 Spectral Filtering in Mamyshev Oscillators

A key element that distinguishes the Mamyshev oscillator from other mode-locking techniques is the use ofoffset spectral filtering. Each round trip contains two bandpass filters whose central wavelengths are intentionally detuned. To simulate this mechanism accurately, a frequency-domain filter model must be incorporated into the round-trip mapping.

**Frequency-domain representation.** Because optical filtering is fundamentally a linear operation, it is most naturally implemented in the frequency domain. Let $A(z,T)$ denote the pulse envelope at position $z$, and let $\tilde{A}(z,\omega)$ be its Fourier transform. A spectral filter with center frequency $\omega_c$ and bandwidth $\sigma$ is described by the transfer function

$$H(\omega) = \exp\left[ -\frac{(\omega-\omega_c)^2}{2\sigma^2} \right], \tag{3-22}$$

which corresponds to an idealized Gaussian bandpass filter.

The effect of the filter on the pulse envelope is given by

$$\tilde{A}_{\text{out}}(\omega) = H(\omega)\tilde{A}_{\text{in}}(\omega). \tag{3-23}$$

Transforming back to the time domain yields

$$A_{\text{out}}(T) = \mathscr{F}^{-1}\left[H(\omega)\tilde{A}_{\text{in}}(\omega)\right], \tag{3-24}$$

where $\mathscr{F}^{-1}$ denotes the inverse Fourier transform. In numerical simulations, Eqs. (3-23)–(3-24) are implemented after each fiber segment described by the GNLSE.

## 3.6 Split-Step Fourier Method (SSFM)

To numerically integrate the GNLSE, the SSFM offers an efficient operator-splitting framework[20]. The method relies on decomposing the total propagation operator into a linear dispersion operator $\hat{D}$ and a nonlinear operator $\hat{N}$, such that the GNLSE can be written as

$$\frac{\partial A(z,T)}{\partial z} = \hat{D}A + \hat{N}A. \tag{3-25}$$

The exact formal solution over a short distance $h$ is

$$A(z+h,T) = \exp\left[h(\hat{D}+\hat{N})\right]A(z,T), \tag{3-26}$$

but the two operators generally do not commute, making direct evaluation intractable. SSFM approximates the evolution by treating dispersion and nonlinearity independently within each step. In its symmetric form, the evolution operator is written as

$$\exp\left[h(\hat{D}+\hat{N})\right] \simeq \exp\left(\frac{h}{2}\hat{N}\right)\exp\left(h\hat{D}\right)\exp\left(\frac{h}{2}\hat{N}\right) + \mathscr{O}(h^3), \tag{3-27}$$

which provides improved accuracy while preserving numerical stability.

As shown in Fig. 3-1, the nonlinear operator is evaluated in the time domain because Kerr nonlinearity depends directly on the instantaneous field intensity. Over half-step $h/2$, the nonlinear update is

$$A(T) \to A(T)\exp\left[i\gamma|A(T)|^2\frac{h}{2}\right]. \tag{3-28}$$

The dispersion operator, by contrast, is most naturally applied in the frequency domain. After performing a Fourier transform, the field spectrum evolves as

$$\tilde{A}(\omega) \to \tilde{A}(\omega)\exp\left[-i\left(\frac{\beta_2}{2}\omega^2 + \frac{\beta_3}{6}\omega^3 + \cdots\right)h\right]. \tag{3-29}$$

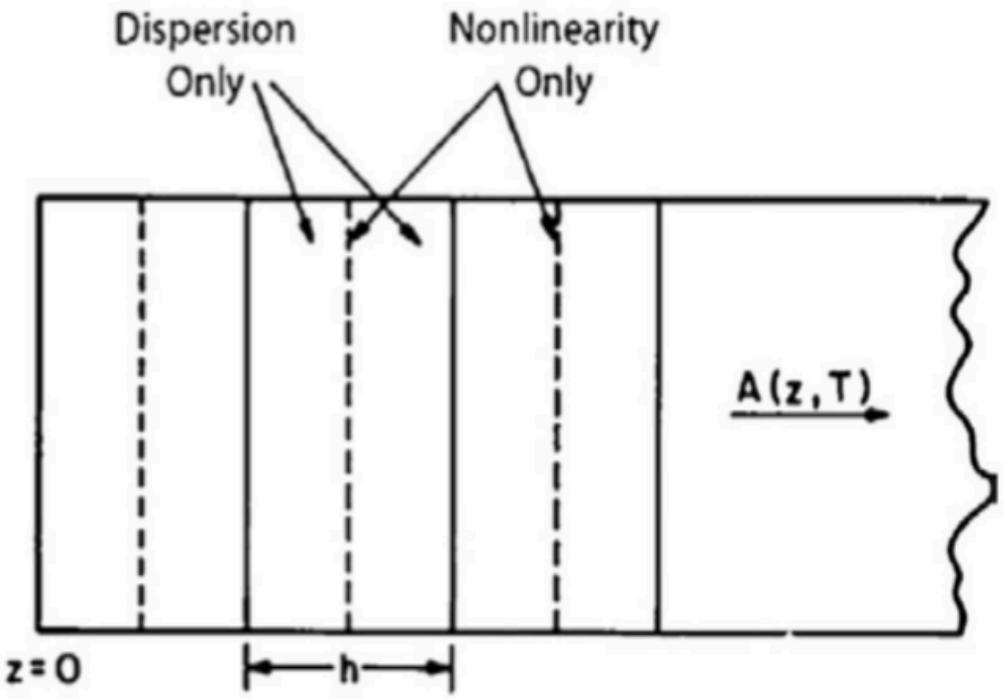



Figure 3-1 Conceptual representation of SSFM[21].

Finally, the field is transformed back to the time domain, and the nonlinear operator is applied again over the remaining half-step. This sequence constitutes one complete SSFM propagation step.

By alternating between dispersion and nonlinearity in this manner, SSFM captures ultrafast pulse evolution while maintaining excellent computational efficiency. As long as the step size $h$ remains sufficiently smaller than the characteristic dispersion and nonlinear lengths of the fiber, the method provides accurate and stable simulations for Mamyshev oscillators.

## 3.7 Runge–Kutta in the Interaction Picture (RK4IP)

While SSFM offers a simple and efficient operator-splitting method, its accuracy can deteriorate when the nonlinear and dispersive effects interact strongly, or when higher-order terms in the GNLSE become important. To improve accuracy without significantly increasing computational cost, the RK4IP provides a powerful alternative[22]. The method naturally incorporates the linear dispersive operator exactly while treating the nonlinear operator with a fourth-order Runge–Kutta scheme.

The generalized nonlinear Schrödinger equation can be written in operator form as

$$\frac{\partial A(z,T)}{\partial z} = \hat{D}A(z,T) + \hat{N}[A(z,T)], \tag{3-30}$$

where $\hat{D}$ is the linear dispersion operator and $\hat{N}$ contains all nonlinear terms such as the Kerr effect, Raman response, and gain saturation.

To construct the interaction picture, the field envelope is transformed as

$$A(z,T) = \exp\left(z\hat{D}\right) U(z,T), \tag{3-31}$$

so that $U(z,T)$ evolves only under the nonlinear operator. Substitution into the GNLSE yields the interaction-picture evolution equation

$$\frac{\partial U}{\partial z} = \exp\left(-z\hat{D}\right) \hat{N} \left[\exp\left(z\hat{D}\right) U\right]. \tag{3-32}$$

This formulation isolates the nonlinear dynamics while treating the dispersive effects analytically through exponential operators.

Over a propagation step $h$, the field in the interaction picture evolves according to a fourth-order Runge–Kutta update:

$$k_1 = F(U), \tag{3-33}$$

$$k_2 = F\left(U + \frac{h}{2}k_1\right), \tag{3-34}$$

$$k_3 = F\left(U + \frac{h}{2}k_2\right), \tag{3-35}$$

$$k_4 = F(U + hk_3), \tag{3-36}$$

where the nonlinear operator in the interaction picture is defined as

$$F(U) = \exp\left(-z\hat{D}\right) \hat{N} \left[\exp\left(z\hat{D}\right) U\right]. \tag{3-37}$$

The updated interaction-picture field is then

$$U(z+h,T) = U(z,T) + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \tag{3-38}$$

Finally, the physical field envelope is recovered by transforming back into the original picture:

$$A(z+h,T) = \exp\left[(z+h)\hat{D}\right] U(z+h,T). \tag{3-39}$$

RK4IP retains excellent accuracy even in the presence of strong nonlinearities or broadband spectral evolution, both of which are characteristic of Mamyshev oscillators.

# 4 Simulation and Analysis of Mamyshev Oscillator Structures

In this chapter, we conduct numerical simulations of the ring-cavity and linear-cavity Mamyshev Oscillator implemented in MATLAB. By varying key parameters such as the filter bandwidth, the spectral offset between the two filters, the pump power, and the length of the gain fiber (GF) section, we investigate their influence on the resulting pulse characteristics. Through this analysis, we establish the relationships between the parameters and characteristics of the output pulse.

## 4.1 Simulation and Analysis of the Ring-Cavity Mamyshev Oscillator

As shown in Fig. 4-1, the constructed laser cavity operates as follows. The light generated by the pump source first passes through the combiner and then propagates through the Yb-doped fiber. At the first filter, the pulse undergoes the first spectral selection and a portion of the light is extracted as the output. The remaining light continues to propagate through the passive fiber (PF) and is further filtered by the second filter. In the numerical simulations, a Gaussian pulse with a temporal width of 1 ps and an initial energy of 1 nJ was used as the input field. This choice ensures that the noise components in the frequency domain are sufficient to trigger mode locking, while avoiding the excessive gain perturbation that would result from a higher-energy seed pulse.

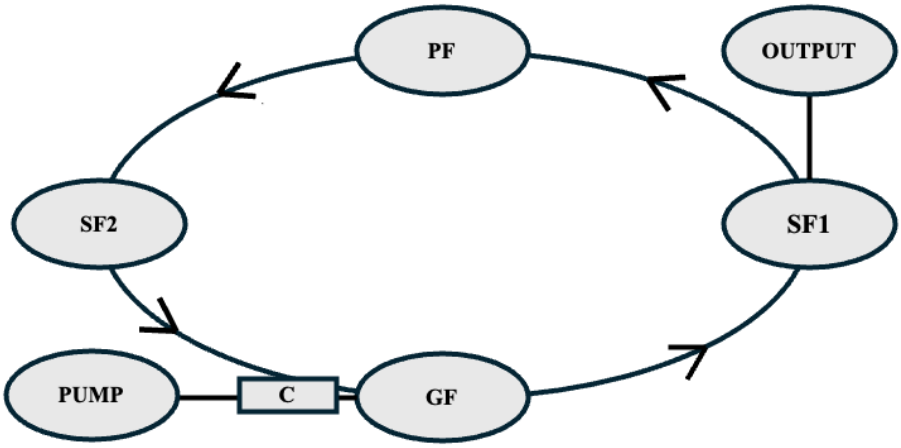



Figure 4-1 Schematic of the all-fiber Mamyshev oscillator. C, combiner; GF, gain fiber; SF, spectral filter; COL, collimator; PF, passive fiber.

### 4.1.1 Pulse Evolution and Output Characteristics in the Oscillator

To investigate the factors influencing the performance of the Mamyshev laser, the parameter designs of experimentally demonstrated Mamyshev mode-locked lasers were first referred to in order to determine an appropriate parameter range for this study. The stability region of the cavity design was then explored by repeatedly adjusting different parameter combinations. A representative set of parameters corresponding to one stable solution obtained from the simulations is summarized in Table 4-1.

Table 4-1 Key parameters of the ring-cavity Mamyshev oscillator.

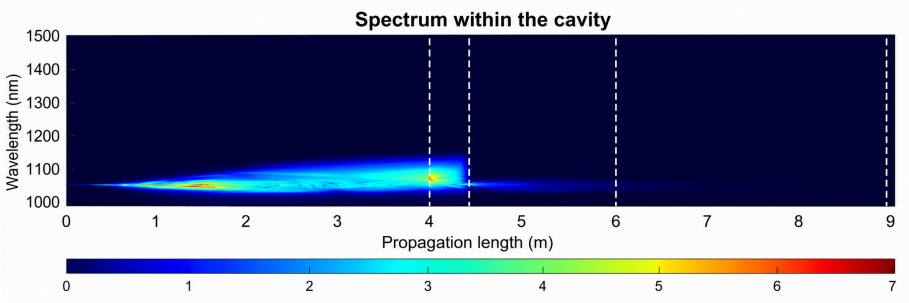
| Category | Parameter | Value |
|---|---|---|
| Passive fiber | MFD | 10.5 $\mu$m |
| | Length | 0.4, 1.5, 3.2 m |
| Gain fiber | Central wavelength $\lambda_0$ | 1035 nm |
| | MFD | 11.1 $\mu$m |
| | Length | 4.0 m |
| | Core diameter | 10 $\mu$m |
| | Cladding diameter | 125 $\mu$m |
| | Cladding absorption at 975 nm | 6.4 dB/m |
| | NA | 0.08 |
| | Pump wavelength | 976 nm |
| | Pump power | 4 W |
| Filter | SF1 | 1040 nm / 5 nm |
| | SF2 | 1030 nm / 3 nm |
| Cavity | Output coupling | 0.9 |

Based on the resonator configuration shown in Fig. 4-1, a numerical model of the Mamyshev oscillator was implemented in MATLAB. The initial pulse was chosen as a normalized Gaussian pulse with a central wavelength of 1035 nm. During each round trip, the pulse sequentially propagates through the gain fiber (GF), the first spectral filter (SF1), the passive fiber (PF), and the second spectral filter (SF2). The parameters used in the simulation are summarized in Table 4-1. Using these parameters, the pulse evolution was computed by iteratively solving the propagation equation over cavity round trips.

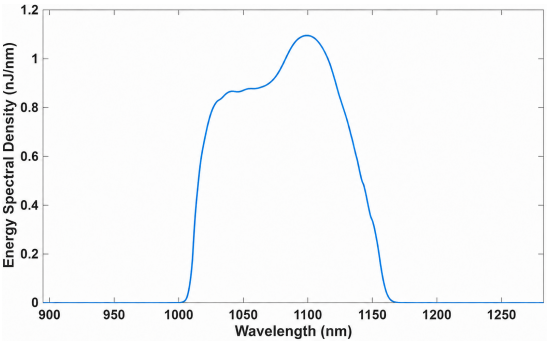
Figure 4-2(a) presents the spectral evolution of the pulse over one round trip within the

cavity. As the pulse propagates through the gain fiber and passive fiber segments, its spectrum broadening due to SPM. Upon reaching the first filter, it removes the low-intensity spectral, while the second filter further narrows and reshapes the spectrum. After filtering, only the strongly broadened central portion of the spectrum is retained, which seeds the next round trip and enables stable, self-consistent mode locking.

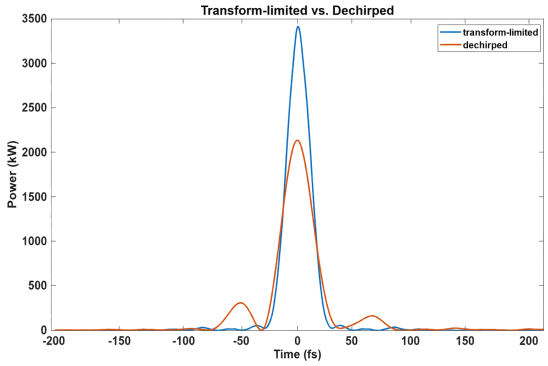
Figure 4-2(b) shows the output spectrum of the Mamyshev oscillator after reaching a stable mode-locking state. The right side of this spectrum is broadened to 1150 nm, indicating significant nonlinear phase accumulation and efficient spectral broadening. Figure 4-2(c) shows the transform-limited pulse and the numerical dechirped pulse. The dechirped pulse has a peak power of approximately 2 MW with a duration of around 40 fs. Small side pulses appear due to residual phase distortion and the asymmetric spectrum.




(a) Spectrum evolution within the cavity.




(b) Output spectrum.




(c) Transform-limited vs. dechirped pulse.

Figure 4-2 (a) Spectral evolution inside the cavity; (b) steady-state output spectrum; (c) comparison between transform-limited and dechirped pulse in the time domain.

During numerical simulation of the stable mode-locking solution, we observed that many cavity parameters have significant influence on the pulse characteristics. Only when these parameters fall within a suitable range can stable mode-locking be realized. Under actual experimental conditions, usually the types of single-mode fiber and Yb-doped gain fiber are fixed,

and at this time, the main factors determining the pulse characteristics are the bandwidths of the two offset spectral filters, their center wavelength interval, pump power, and the length of the Yb-doped gain fiber. This paper utilizes the developed simulation code to systematically study how the above parameters affect the pulse output.

### 4.1.2 Influence of Pump Power on Pulse Characteristics

To investigate the impact of pump power on the output pulse characteristics, all other parameters of the Mamyshev oscillator were held constant. The fixed parameters used in this study are summarized in Table 4-1. The pump power was swept from 3 W to 7 W in steps of 0.5 W. For each pump value, numerical simulations were performed in MATLAB based on the cavity configuration illustrated in Fig. 4-1. The pulse waveform at the output port of the cavity was extracted after the steady-state mode-locked regime was reached. Subsequently, the corresponding pulse energy, peak power and full width at half maximum FWHM were computed. The resulting dependencies of these characteristics on the pump power are presented in Fig. 4-3, 4-4.

Figure 4-3 shows the variation of pulse energy and peak power as the pump power increases from 3 W to 7 W. Both quantities exhibit an obvious monotonic increase, reflecting the gain enhancement provided by the Yb-doped fiber under high pump power. When the pump power exceeds 6 W, the curves begin to saturate, indicating that the gain medium is approaching partial saturation, and the power growth brought by increasing the pump power tends to zero.

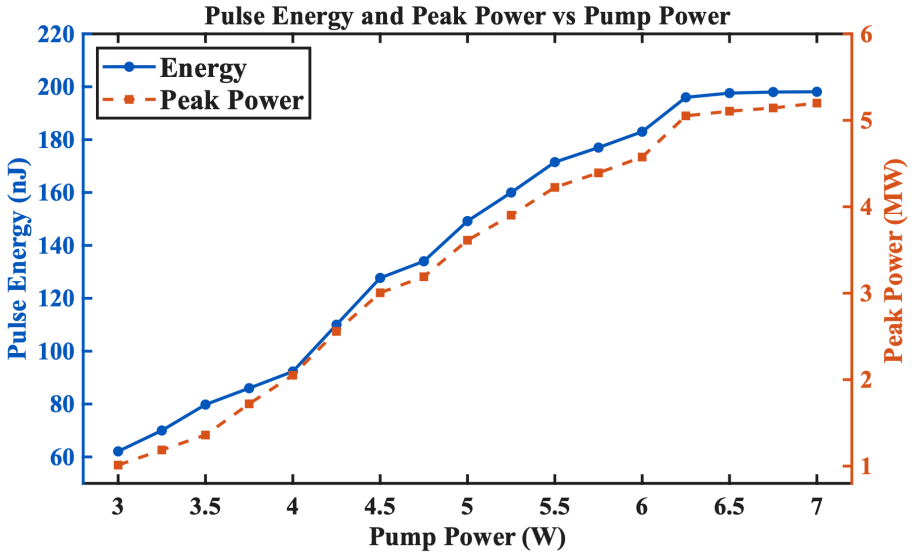



Figure 4-3 Pulse energy and peak power versus pump power.

Figure 4-4 illustrates the numerical simulation results for the evolution of output pulse width as a function of pump power. It can be observed that the pulse width decreases rapidly from approximately 61 fs at 3 W to about 45 fs at 4 W, after which the rate of decline slows, approaching a slight saturation value of nearly 38 fs at 7 W. Within the range where pump power increases from 3.5 W to 4 W, the pulse width drops significantly. This occurs because, once the spectral broadening induced by SPM crosses the critical threshold of the filter offset, the spectral components intercepted by the filter expand rapidly from the edges to cover the center of the passband. This leads to a step-like growth in bandwidth, which in turn triggers a sharp contraction of the pulse width. As the pump power further increases beyond 6 W, the pulse width enters a saturation plateau at approximately 38 fs. This limitation primarily stems from two factors. First, the filter passband bandwidth restricts any further growth in spectral width. In the numerical simulations, it was observed that although the raw spectrum inside the fiber continues to broaden aggressively with increasing nonlinearity, the output spectral width after the filter remains constant. This indicates that the 5 nm filter passband limits the coherent bandwidth available for temporal compression; the 38 fs lower limit is highly consistent with the transform-limited width corresponding to this bandwidth. Second, the intense SPM at extremely high power levels introduces complex high-order nonlinear phases and nonlinear chirp while contributing to spectral broadening. Such high-order phase distortions are difficult to eliminate through conventional second-order dispersion compensation. The accumulation of residual phase distortions within the pulse ultimately limits the further contraction of the main pulse peak.

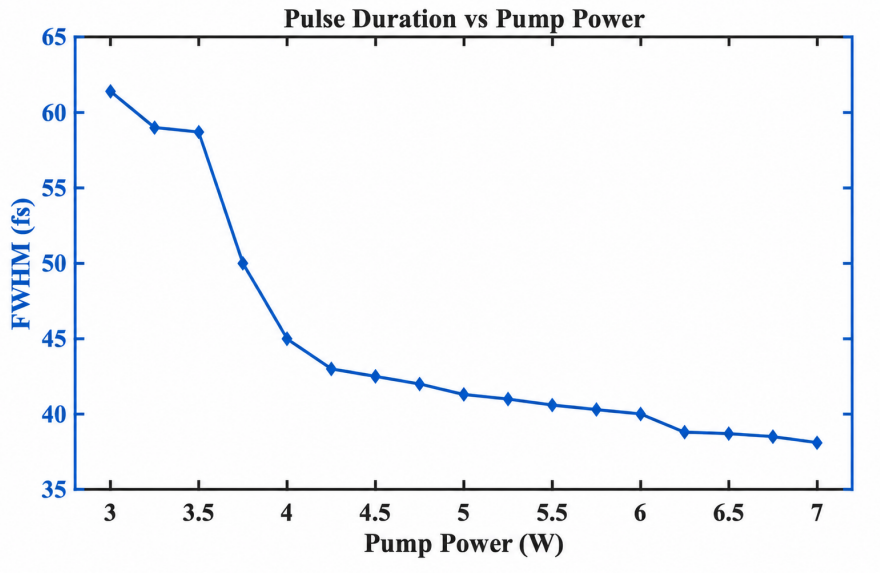



Figure 4-4 Pulse duration versus pump power.

In addition, when the pump power exceeds 7 W, the cavity can no longer sustain a stable single-pulse mode-locked state, indicating that excessive pump levels drive the system into a multi-pulse or unstable regime.

### 4.1.3 Influence of filter offset on Pulse Characteristics

Next, the influence of the spectral offset between the two filters on the output pulse characteristics is investigated. The center wavelength of SF1 was fixed at 1040 nm, and the filter offset was adjusted by changing the center wavelength of SF2, while the other parameters were kept the same as those listed in Table 4-1. The offset was scanned from 4 nm to 12 nm with a step of 1 nm, so that the influence of different spectral offsets on the final output pulse could be investigated. For each offset condition, numerical simulations were performed, and the corresponding pulse energy, peak power, pulse duration, and output center wavelength were obtained.

Fig. 4-5 demonstrates the variation of pulse energy and peak power in the process of the filter interval increasing from 4 nm to 12 nm. It can be seen that when the interval increases from 4 nm to about 10 nm, the pulse energy shows an obvious downward trend, decreasing from about 110 nJ to about 92 nJ. This is because a larger interval requires the pulse to generate stronger SPM in the fiber to realize sufficient spectral broadening, so that it can successfully pass through the passbands of the two filters. Insufficient broadening will cause more spectrum to be filtered out, thereby increasing the intracavity loss and causing energy drop. At the same time, the peak power rises slowly with the increase of the interval. This mainly originates from the pulse becoming shorter (see Fig. 4-6), that is, under the condition of slightly reduced energy, due to the significant compression of pulse width, the peak power increases accordingly. When the interval further increases to above 10 nm, both pulse energy and peak power rebound, indicating that the pulse has already accumulated sufficiently strong nonlinear broadening, and at this time, the filtering loss is alleviated, and the output energy is partially recovered.

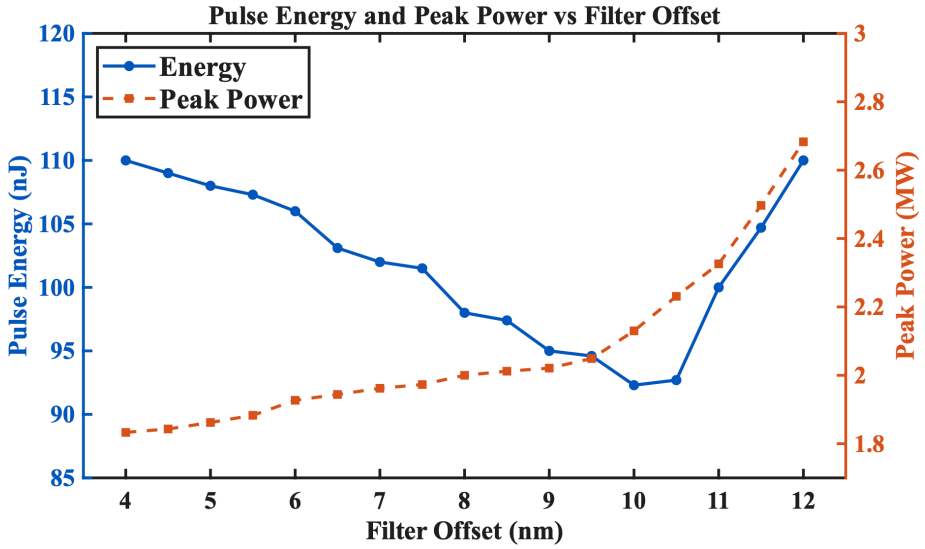



Figure 4-5 Pulse energy and peak power versus filter offset.

Fig. 4-6 shows that the pulse width decreases monotonically as the filter interval increases, reducing from about 60 fs to about 41 fs. As the interval between the passbands of the two filters increases, the system has higher requirements for spectral broadening, and only the spectral parts broadened by stronger SPM can pass through the filters. This more strict filtering window makes the output pulse tend to be a short pulse closer to the transform limit, so the pulse width presents a trend of continuous compression.

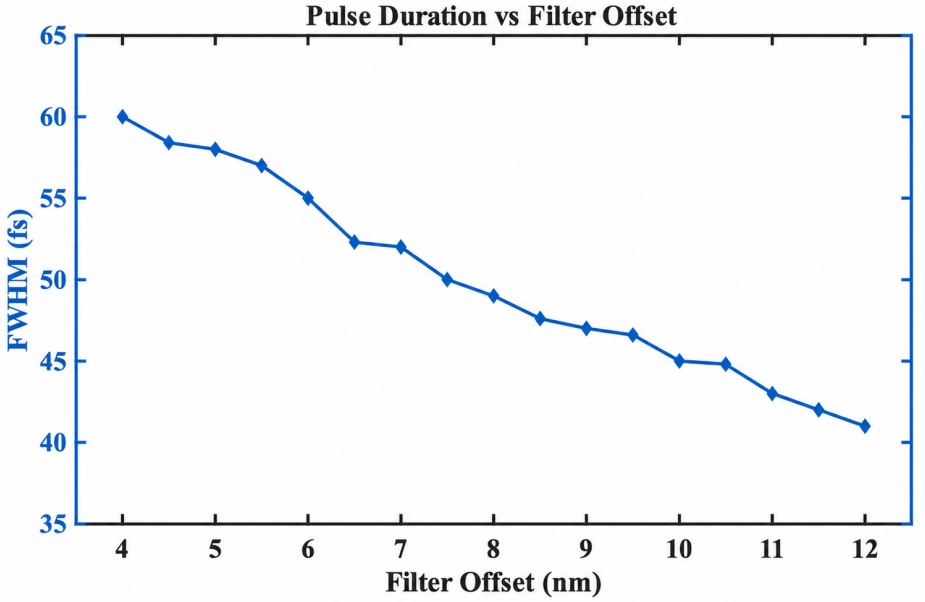



Figure 4-6 Pulse duration versus filter offset.

Fig. 4-7 demonstrates the variation law of the output center wavelength as the filter inter-

val increases. In the process of the offset increasing from 4 nm to 12 nm, the center wavelength continuously blue-shifts from about 1035.7 nm to about 1030.1 nm, presenting a stable monotonic decrease. This blue-shift mainly originates from the passable wavelength range of the pulse shifting towards the short-wavelength direction with the increase of the offset. When the distance between the center wavelengths of the two filters becomes larger, their overlapping effective passband will move to the shorter wavelength side, thereby leading to the blue-shift of the output wavelength.

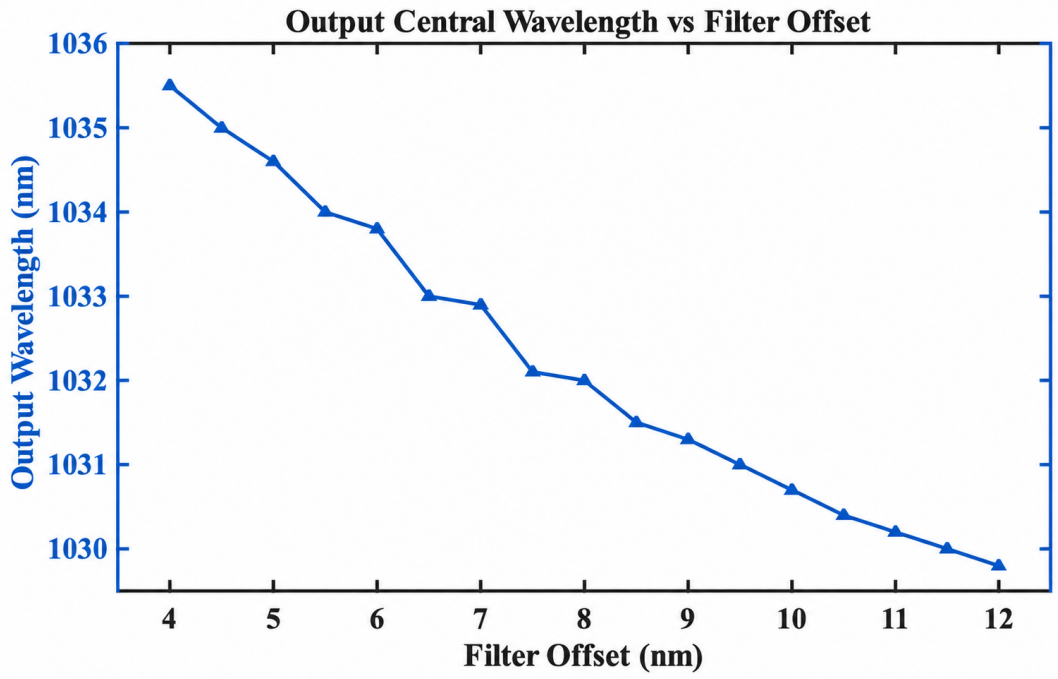



Figure 4-7 Center wavelength versus filter offset.

During the simulation, an additional offset-scanning scheme was also tested, in which the center wavelength of SF2 was fixed and the center wavelength of SF1 was varied. The obtained results were consistent with those obtained by adjusting SF2. Therefore, only the results based on SF2 tuning are presented in this section.

### 4.1.4 Influence of filter bandwidth on Pulse Characteristics

The center wavelength and bandwidth of SF1, as well as the other cavity parameters, were kept fixed, and the bandwidth of SF2 was varied to investigate its influence on the pulse output characteristics. The bandwidth takes values in the interval of 1-6 nm with an interval of 1 nm, and the pulse output is obtained at the output position in the resonant cavity structure shown

in Fig. 4-1. Through numerical simulation of each offset condition, the corresponding pulse energy, peak power and pulse duration are obtained.

From Fig. 4-8, it can be seen that as the bandwidth increases from 1 nm to 3 nm, both show a significant upward trend: the pulse energy increases from about 80 nJ to about 92 nJ, and the peak power also increases from about 1.55 MW to over 2.0 MW. This is because a wider bandwidth allows more spectrally broadened components after nonlinear broadening to pass through the filter, thereby reducing spectral loss, making the pulse retain higher energy, and generating a greater nonlinear compression effect. When the bandwidth exceeds 3 nm, both energy and peak power begin to decrease. It can be seen that, under the condition that other parameters in this laser resonant cavity structure are fixed, there exists an optimal solution for the offset filter bandwidth. Within the selection range of offset filter bandwidth that can realize stable mode-locking output, when the offset filter bandwidth is smaller than the optimal solution, the pulse power increases with the increase of the offset filter bandwidth; when the offset filter bandwidth is larger than the optimal solution, the pulse power decreases with the increase of the offset filter bandwidth.

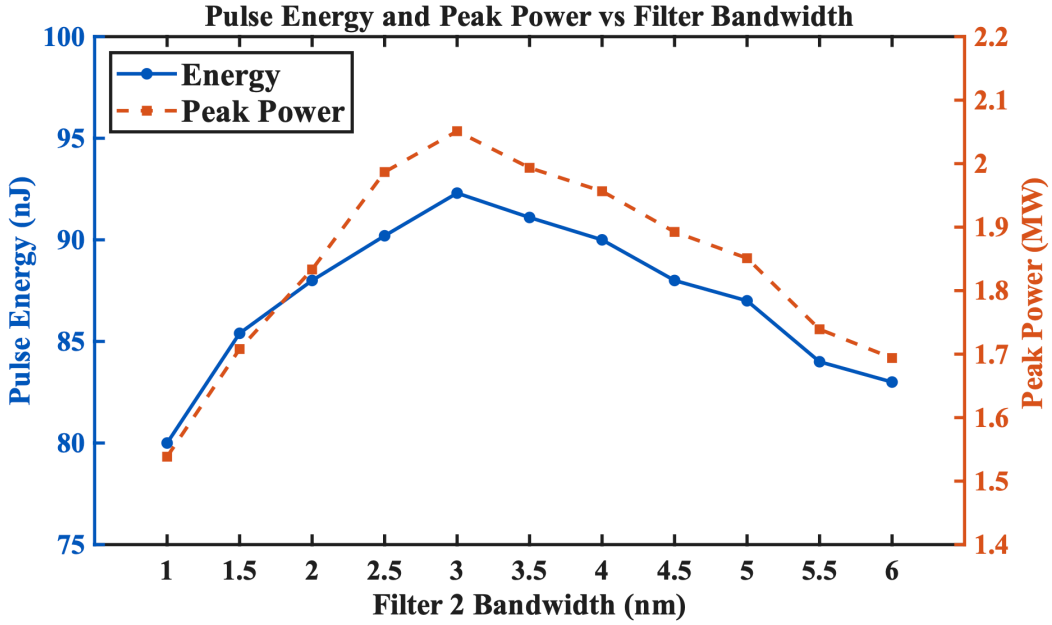



Figure 4-8 Pulse energy and peak power versus bandwidth.

Fig. 4-9 shows that under the condition that other parameters we selected are fixed, as the offset filter bandwidth increases from 2 nm to 5 nm, the duration of the pulse is relatively close. But in the process of simulation, we found that when the bandwidth is 1 nm, the pulse can realize stable circulation inside the cavity, but the output spectrum is relatively messy. This is

because the frequency components allowed to pass by the offset filter are relatively few, and the pulse after being filtered passes through the single-mode fiber and gain fiber, unable to realize stable broadening under the effects of gain, dispersion, and nonlinearity. When the offset filter bandwidth is 6 nm, as the pulse circulates inside the cavity, its shape becomes more and more messy, and finally unable to form stable output. This is because the frequency components allowed to pass by the offset filter are relatively many, and the part of the pulse retained after passing through the filter passes through the single-mode fiber and gain fiber, unable to reach balance under the effects of gain, dispersion, and nonlinearity.

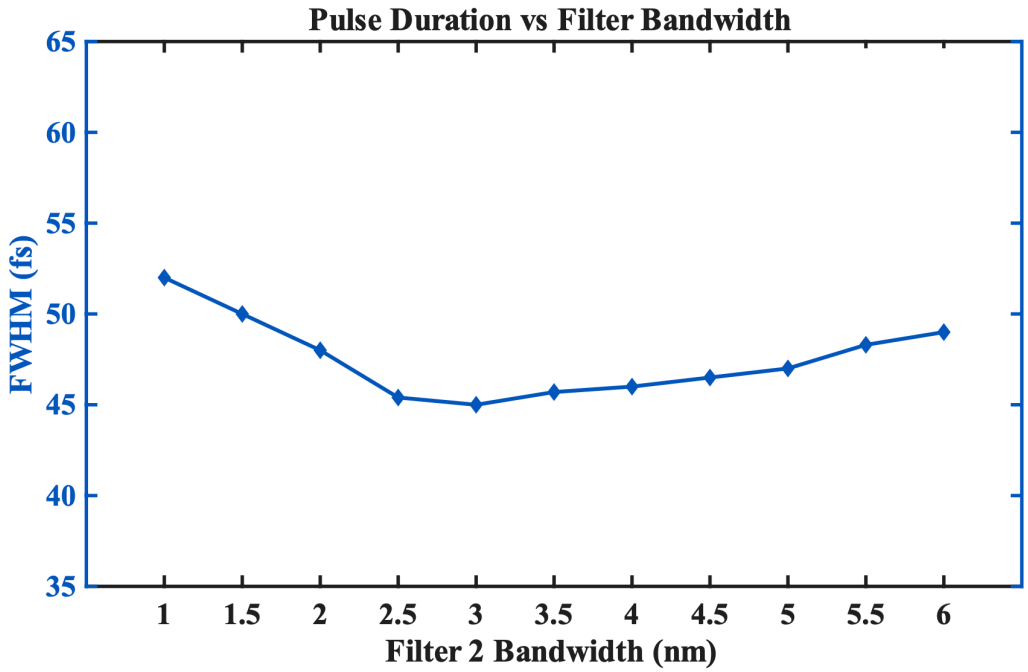



Figure 4-9 Pulse duration versus filter offset.

Similarly, an additional bandwidth-scanning scheme was tested during the simulation, in which the bandwidth of SF2 was fixed while the bandwidth of SF1 was varied. The obtained results were consistent with those obtained by varying the bandwidth of SF2. When the bandwidths of SF1 and SF2 were varied simultaneously, significant fluctuations were observed in the output characteristics, and no clear regular dependence could be identified.

### 4.1.5 Influence of Gain Fiber Length on Pulse Characteristics

The other parameters were kept unchanged, and the length of the Yb-doped gain fiber was varied to investigate its influence on the pulse output characteristics. The gain fiber length was set to 2.5, 3.2, 4.0, 4.8, and 5.6 m, respectively. To maintain the balance between gain accumulation, dispersion, and nonlinear phase accumulation, each passive fiber section was scaled proportionally with the gain fiber length. This proportional adjustment ensured that the cavity could still converge to a stable single-pulse solution under different gain-fiber-length conditions.

Fig. 4-10 illustrates the trends of output pulse energy and peak power as a function of gain fiber length. The results show that both parameters exhibit a steady upward trend, with the pulse energy increasing from approximately 70 nJ to over 110 nJ. The direct driver for this phenomenon is that a longer gain fiber significantly enhances the total effective absorption of pump light, allowing the pulse to accumulate more gain during each round trip. At a deeper level, the increase of fiber length extends the nonlinear interaction distance during amplification, enabling the pulse to accumulate stronger SPM effects alongside energy gain. This simultaneous intensification of gain and nonlinearity not only boosts the energy directly but also results in broader spectral broadening, which facilitates pulse width contraction in the time domain through the subsequent filtering process. Comparison reveals that the growth slope of the peak power is slightly higher than that of the pulse energy; this is a direct consequence of the combined effect of energy enhancement and pulse width contraction (see Figure 4-10). Furthermore, within this range of fiber lengths, the energy curve shows no obvious slowdown, suggesting that the system has not yet reached the gain saturation bottleneck or the threshold for severe nonlinear instability. The increase in gain fiber length effectively optimizes the balance between energy extraction efficiency and the intensity of nonlinear evolution.

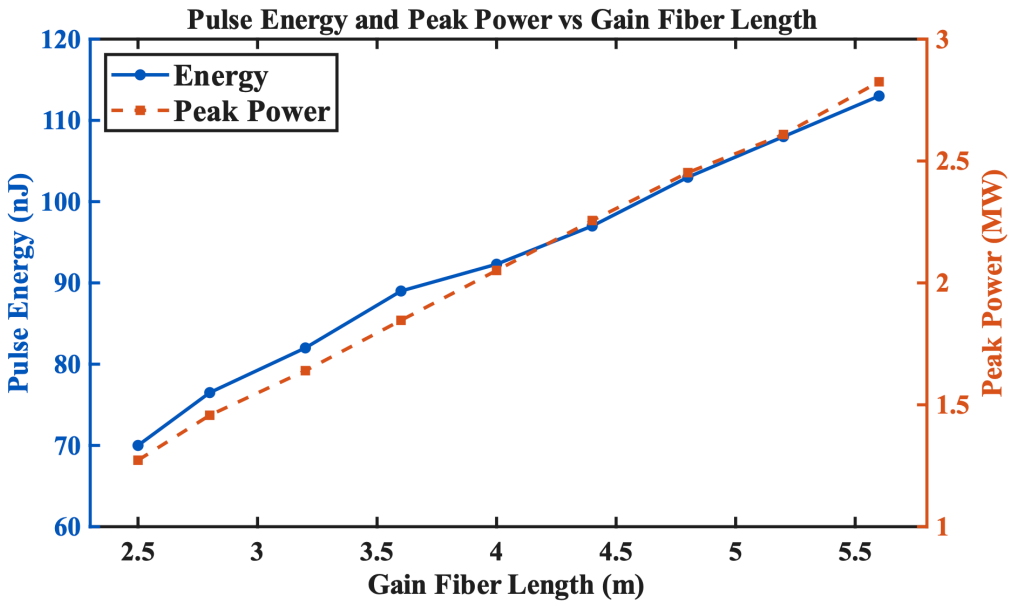



Figure 4-10 Pulse energy and peak power versus gain fiber length.

From Fig. 4-11, it can be seen that the pulse duration decreases from about 55 fs to about 40 fs. As the gain fiber length becomes longer, the pulse experiences stronger SPM in the fiber, thereby generating more obvious spectral broadening. Under the subsequent action of the filter, the broadened spectrum is effectively compressed, making the output pulse tend to become more short-pulsed.

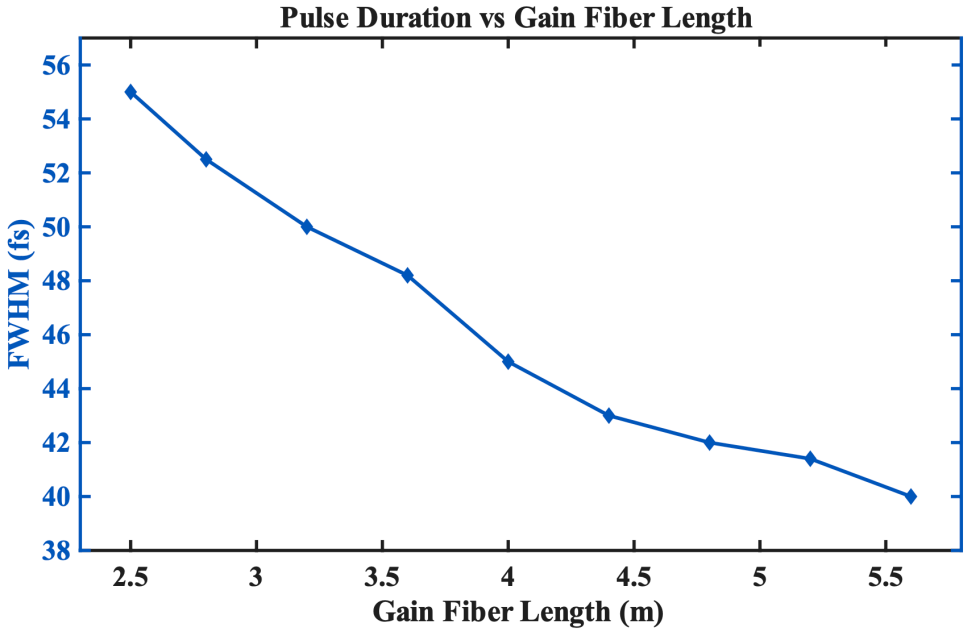



Figure 4-11 Pulse duration versus gain fiber length.

## 4.2 Simulation and Analysis of linear-cavity Mamyshev Oscillator

As shown in Fig. 4-12, the constructed laser cavity operates as follows. The light generated by the pump source first passes through the combiner and then propagates through the Yb-doped fiber. At the first fiber Bragg grating, the pulse undergoes the first spectral selection. The selected light enter the linear cavity and propagate through the passive fiber (PF) and the gain fiber. Then further filtered by the second fiber Bragg grating. Only the high intensity light can reflect back to the cavity and undergoes the second spectral selection. The light that reflects back to the first fiber Bragg Grating is chosen as the monitoring output. In the numerical simulations, a Gaussian pulse with a temporal width of 1 ps and an initial energy of 1 nJ was used as the input field.

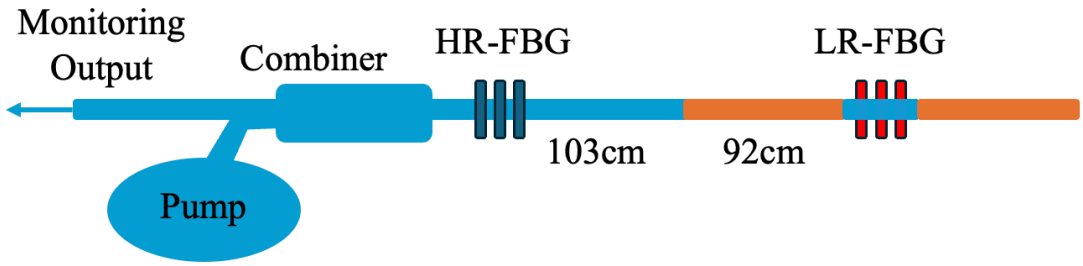



Figure 4-12 Schematic of the all-fiber linear-cavity Mamyshev oscillator. The blue/orange sections are Coractive passive 10/125DC-PM and gain DCF-YB-10/125E-PM fibers, respectively. HR/LR-FBG, high/low reflectivity fiber Bragg gratings.

### 4.2.1 Pulse Evolution and Output Characteristics in the Oscillator

Similarly to the ring-cavity, The approximate parameter region is observed to further explore the influence of different parameters. A representative set of parameters corresponding to one stable solution obtained from the simulations is summarized in Table 4-2.

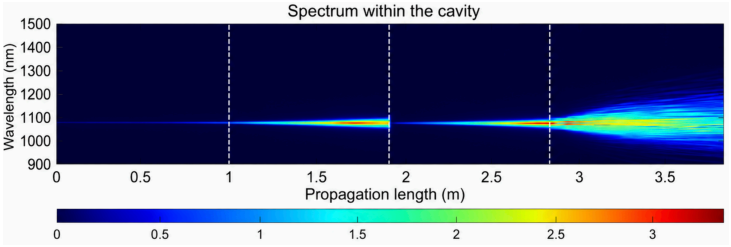
Table 4-2 Key parameters of the linear-cavity Mamyshev oscillator.

| Category | Parameter | Value |
|---|---|---|
| Passive fiber | MFD | 10.5 $\mu$m |
| | Length | 1.03 m |
| Gain fiber | Central wavelength $\lambda_0$ | 1033 nm |
| | MFD | 11.1 $\mu$m |
| | Length | 0.92 m |
| | Core diameter | 10 $\mu$m |
| | Cladding diameter | 125 $\mu$m |
| | Cladding absorption at 975 nm | 6.4 dB/m |
| | NA | 0.08 |
| | Pump wavelength | 976 nm |
| | Pump power | 4 W |
| FBG | HR-FBG | 1030 nm / 1.5 nm, $R = 55\%$ |
| | LR-FBG | 1036 nm / 1.5 nm, $R = 15\%$ |
| Cavity | Output coupling | 0.9 |

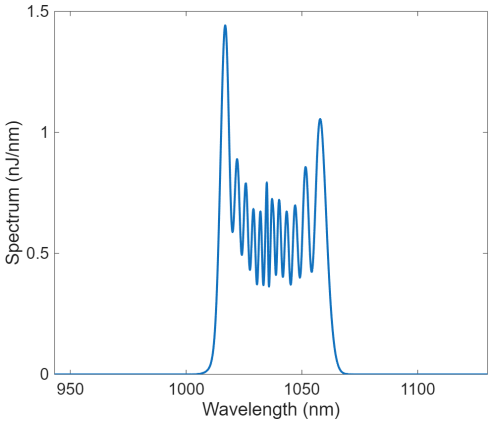
Based on the resonator configuration shown in Fig. 4-12, a numerical model of the Mamyshev oscillator was implemented in MATLAB. The initial pulse was chosen as a normalized Gaussian pulse with a central wavelength of 1033 nm.

Figure 4-13(a) presents the spectral evolution of the pulse over one round trip within the cavity. In the simulation, the pulse travels back and forward in the cavity is set as a single round trip. As the pulse propagates through the gain fiber and passive fiber segments, its spectrum broadening due to SPM. Upon reaching the LR-FBG, it removes the low-intensity spectral and reflect the rest back to the cavity. Same thing happens when the pulse reaches the HR-FBG. After filtering, only the strongly broadened central portion of the spectrum is retained, which seeds the next round trip and enables stable, self-consistent mode locking. Figure 4-13(b) shows the output spectrum of the Mamyshev oscillator after reaching a stable mode-locking state. The right side of this spectrum is broadened to 1070 nm, indicating significant nonlinear phase accumulation and efficient spectral broadening. Figure 4-13(c) shows the transform-limited pulse and the numerical dechirped pulse. The dechirped pulse has a peak power of approximately 400 kW with a duration of around 80 fs. Small side pulses appear due to residual

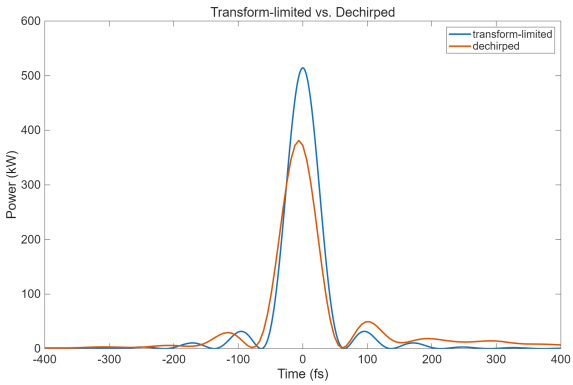
phase distortion and the asymmetric spectrum.




(a) Spectrum evolution within the cavity.




(b) Output spectrum.




(c) Transform-limited vs. dechirped pulse.

Figure 4-13 (a) Spectral evolution inside the cavity; (b) steady-state output spectrum; (c) comparison between transform-limited and dechirped pulse in the time domain.

Similarly to the ring-cavity, we also choose bandwidth of the two FBG, their center wavelength interval, pump power and the length of the Yb-doped gain fiber as the parameters to find out their influences towards the performance of the Mamyshev oscillator.

### 4.2.2 Influence of Pump Power on Pulse Characteristics

To investigate the impact of pump power on the output pulse characteristics, all other parameters of the Mamyshev oscillator were held constant. The fixed parameters used in this study are summarized in Table 4-2. The pump power was swept from 3 W to 7 W in steps of 0.25 W. For each pump value, numerical simulations were performed in MATLAB based on the cavity configuration illustrated in Fig. 4-12. The pulse waveform at the output port of the cavity was extracted after the steady-state mode-locked regime was reached. Subsequently, the corresponding pulse energy, peak power and pulse duration were computed. The resulting dependencies of these characteristics on the pump power are presented in Fig. 4-14 - 4-15.

Figure 4-14 shows the evolution of output pulse energy and peak power as a function of pump power for the linear-cavity Mamyshev oscillator. In the range of 3 W to 7 W, the pulse energy and peak power increase monotonously. The pulse energy growing from approximately 25 nJ to 60 nJ and the peak power rising from 250 kW to over 700 kW. This is primarily because higher pump power provides stronger gain compensation for the pulse during each round trip. Although the curves growth linearly near 7 W, the system can no longer keep a stable single-pulse mode-locked state when the pump power exceed 7 W. Furthermore, the simulation results indicate that under other parameter combinations, the Mamyshev oscillator can sustain stable single-pulse operation at even higher pump levels.

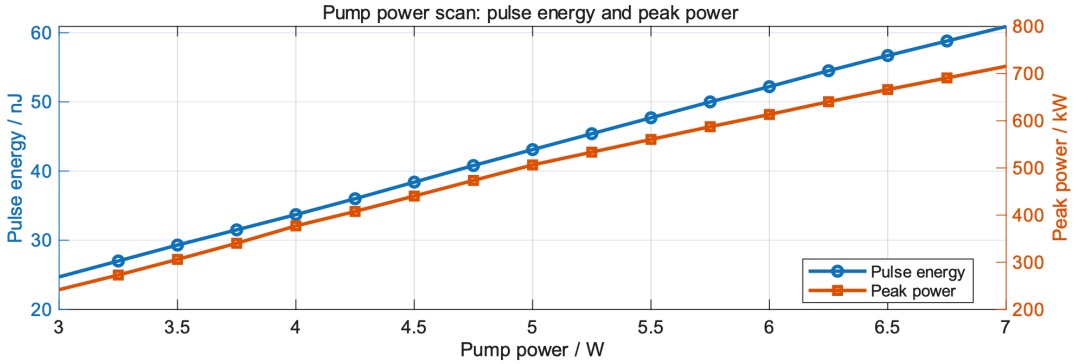



Figure 4-14 Pulse energy and peak power versus pump power.

Figure 4-15 demonstrates the relationship between the pulse duration and the pump power. An obvious decreasing can be observed: the pulse width decreases rapidly from about 96 fs at 3 W to about 80 fs at 5 W and approaches a saturation of 80 fs at 5 W. This phenomenon is due to the fact that under high pump power, the SPM is enhanced and the spectrum broadens more, enabling the spectral filter to generate shorter pulses. The subsequent saturation mechanisms are similar to those of the ring cavity, stemming from the joint constraints of the filter bandwidth and the transform limit. Simulation observations reveal that beyond 5 W, although further increases in pump power continue to drive spectral broadening within the fiber, the output spectral width after FBG reflection tends toward saturation. This indicates that the spectral window of the FBG limits further broadening. Furthermore, under high-power operation, the intense nonlinear chirp induced by strong nonlinearities cannot be fully compensated for by simple gratings. These residual phase distortions ultimately prevent the main pulse peak from narrowing further.

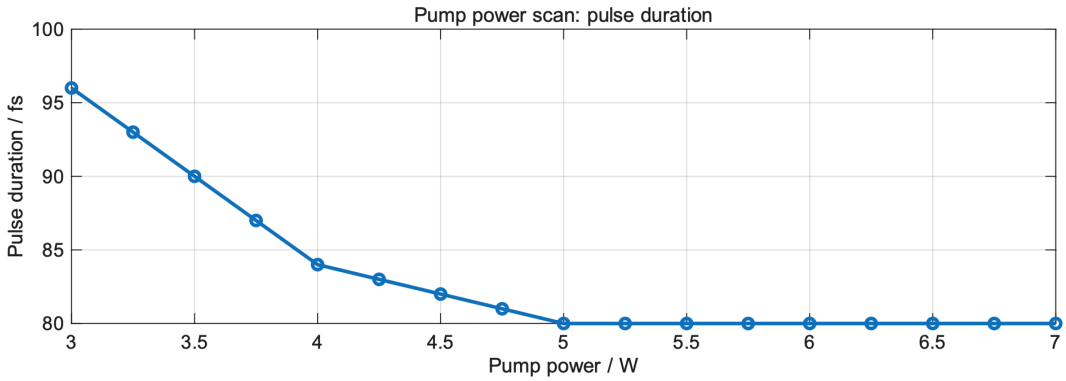



Figure 4-15 Pulse duration versus pump power.

### 4.2.3 Influence of filter offset on Pulse Characteristics

The influence of the spectral offset between the two FBGs on the output pulse characteristics was then investigated. The center wavelength of the HR-FBG was fixed at 1030 nm, and the spectral offset was adjusted by changing the center wavelength of the LR-FBG, while the other parameters were kept the same as those listed in Table 4-2. The offset was scanned from 3 nm to 10 nm with a step of 0.5 nm, to observe how different filter spectral offsets affect the final output pulse.

Fig. 4-16 demonstrates the variation trend of pulse energy and peak power in the process of the filter interval increasing from 3 nm to 10 nm. As the FBG offset increases from 3nm to 10nm, the pulse energy and peak power first rise and then fall, reaching their optimal values around 6-6.5nm. Similar to the principle of the ring cavity, when the offset is small, the spectral interval between the two FBG filtering windows is small, and the pulse does not need to undergo sufficient nonlinear broadening to pass through the FBG, thus selecting a weaker pulse intensity. As the offset increases, the proportion of high-intensity pulses gradually increases, leading to an increase in energy and peak power. However, when the offset continues to increase, the excessive interval between the two FBGs leads to a significant increase in filtering loss, resulting in a decrease in both the final pulse energy and the compressed peak power. This differs from the trend of the ring cavity. Our preliminary analysis suggests that although both of them rely on spectral broadening caused by self-phase modulation, their responses to the filter offset are not entirely consistent. In the ring cavity, the pulse propagates in a single direction, and changes in the filter offset only affect the balance between filtering loss and nonlinear spectral broadening. However, the linear cavity uses a reflective cavity, where the FBG simultaneously serves as both feedback and output coupling. Changing the FBG offset not only

alters the spectral broadening required for Mamyshev filtering, but also changes the feedback spectrum reflected back into the cavity and the transmission output spectrum. Therefore, it is more sensitive to excessive offset and cannot maintain a stable high-intensity pulse under high offset conditions.

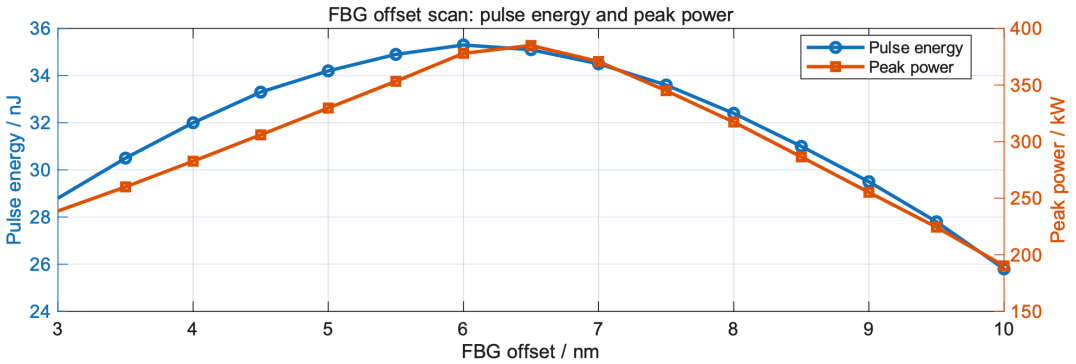



Figure 4-16 Pulse energy and peak power versus filter offset.

Fig. 4-17 illustrates the evolution of the output pulse width as a function of FBG offset for the linear cavity, exhibiting a U-shaped. In the 3 nm to 6.5 nm range, the pulse width shortening results from the enhanced pulse-filtering effect brought by the larger offset. Specifically, a higher offset raises the threshold for the equivalent saturable absorber, forcing the pulse to accumulate stronger SPM to bridge the spectral gap. This mechanism effectively selects narrow pulses with improved linear chirp and higher coherence. However, unlike ring cavities which can typically maintain short pulse output over a broad offset range, the pulse width of the linear cavity rises rapidly once the offset exceeds 6.5 nm. This reflects the different feedback mechanism of the linear configuration: the FBG simultaneously performs the functions of spectral filtering and reflection feedback. When the offset is large, the feedback spectrum reflected back into the cavity is significantly attenuated in energy. This leads to insufficient SPM intensity in the subsequent round-trip, meaning the spectral broadening can no longer support the pulse in spanning the large center-wavelength interval, which in turn causes the pulse width to increase. Consequently, the pulse width performance of the linear cavity exhibits higher sensitivity to the spectral offset.

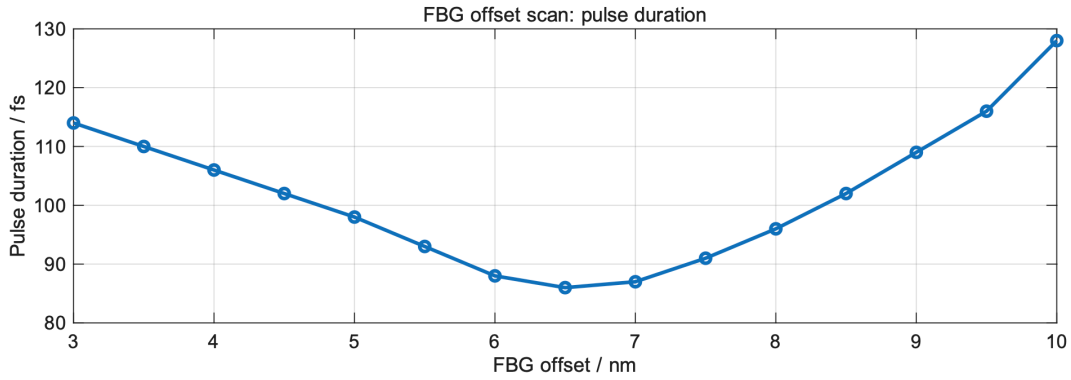



Figure 4-17 Pulse duration versus filter offset.

### 4.2.4 Influence of filter bandwidth on Pulse Characteristics

The center wavelength and bandwidth of the HR-FBG, together with the other cavity parameters, were kept fixed, and the bandwidth of the LR-FBG was varied to investigate its influence on the pulse output characteristics. The bandwidth takes values in the interval of 0.5-4 nm with an interval of 0.25 nm, and the pulse output is obtained at the output position in the resonant cavity structure shown in Fig. 4-12. Through numerical simulation of each offset condition, we obtained the corresponding pulse energy, peak power and pulse duration.

From Fig. 4-18, it can be seen that as the bandwidth increases from 0.5 nm to 2 nm, both show a significant upward trend: the pulse energy increases from about 28 nJ to about 34 nJ, and the peak power also increases from about 225 kW to over 375 kW. When the bandwidth exceeds 2 nm, both energy and peak power begin to decrease. Similar to the ring-cavity, linear-cavity also has an optimal solution. When the offset filter bandwidth is larger than the optimal solution, the pulse power decreases with the increase of the offset filter bandwidth.

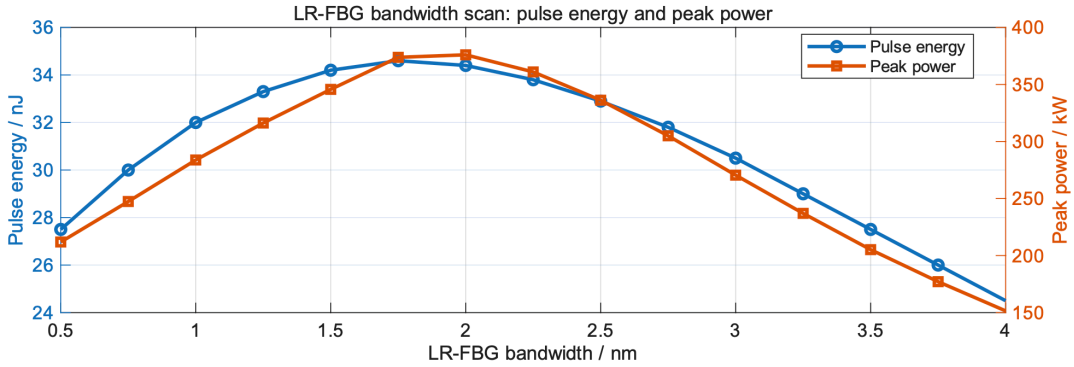



Figure 4-18 Pulse energy and peak power versus bandwidth.

Fig. 4-19 shows that as the offset filter bandwidth increases from 0.5 nm to 4 nm, the duration of the pulse first decrease then increase. Compare to the ring-cavity, the changing is more obvious. This further confirms that linear-cavity is more sensitive to the adjustment of FBG parameters, and that stable pulses within the cavity are more easily disrupted compared to ring-cavity.

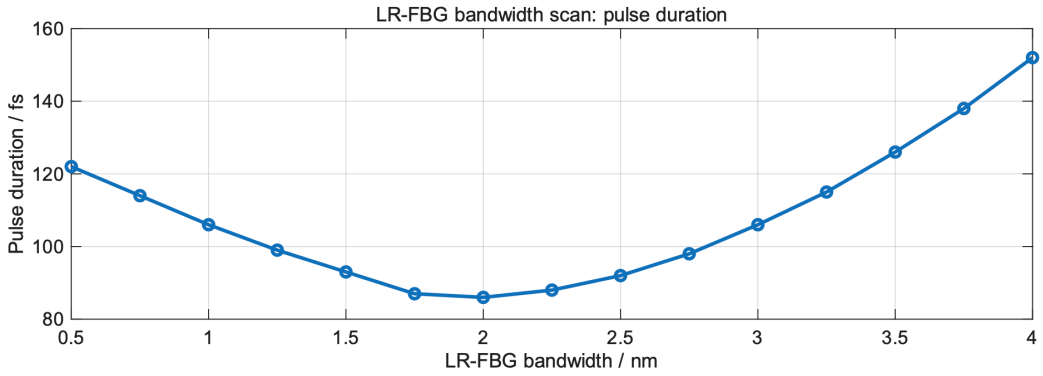



Figure 4-19 Pulse duration versus filter offset.

### 4.2.5 Influence of Gain Fiber Length on Pulse Characteristics

The other cavity parameters were kept unchanged, and the length of the Yb-doped gain fiber was varied to investigate its influence on the pulse output characteristics. The gain fiber length was scanned from 0.7 m to 1.4 m with a step of 0.1 m. To maintain the relative balance among gain accumulation, dispersion, and nonlinear phase accumulation, the passive fiber length was scaled proportionally with the gain fiber length. This adjustment ensured that the cavity could still converge to a stable single-pulse solution under different gain-fiber-length conditions.

Fig. 4-20 illustrates the evolution of pulse energy and peak power as a function of gain fiber length for the linear cavity. A remarkably stable linear upward trend in output performance is observed: the pulse energy grows steadily from an initial 24 nJ to over 55 nJ, while the peak power correspondingly rises from approximately 220 kW to above 750 kW. This originates primarily from the enhanced pump absorption efficiency provided by the longer gain fiber, which allows the pulse to achieve stronger gain compensation during each round trip. Furthermore, the increased fiber length essentially extends the nonlinear interaction distance, facilitating more extensive spectral broadening through enhanced SPM effects. As the curves show no sign of a decreasing slope within this length range, it indicates that the system has not

yet reached the gain saturation bottleneck or encountered severe nonlinear instability.

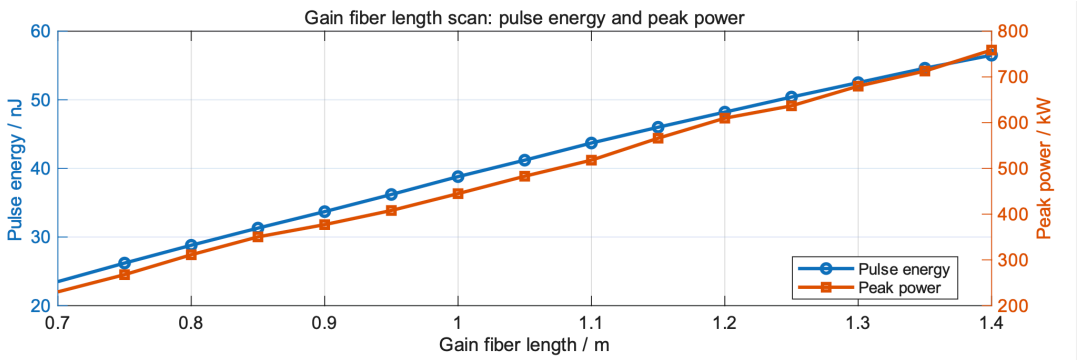



Figure 4-20 Pulse energy and peak power versus gain fiber length.

Fig. 4-21 shows the contraction of the output pulse width as a function of gain fiber length in the linear cavity. As the length extends from 0.7 m to 1.4 m, the pulse width continuously decreases from 96 fs to approximately 70 fs. This originates from the fact that increasing the gain fiber length extends the nonlinear interaction distance of the pulse. In the Mamyshev mechanism, the accumulation of nonlinear effects is directly related to the interaction length; a longer fiber intensifies the SPM effect, thereby driving broader spectral extension. This allows the FBG to intercept a wider spectral bandwidth, resulting in an output pulse that is compressed more narrowly in the time domain. Furthermore, the influence of nonlinear chirp introduced by the added length is not yet sufficient to overshadow the advantages of spectral broadening; consequently, the pulse width exhibits a near-monotonic decreasing trend. This indicates that within this parameter range, shorter ultrashort pulses can be achieved by appropriately increasing the gain length.

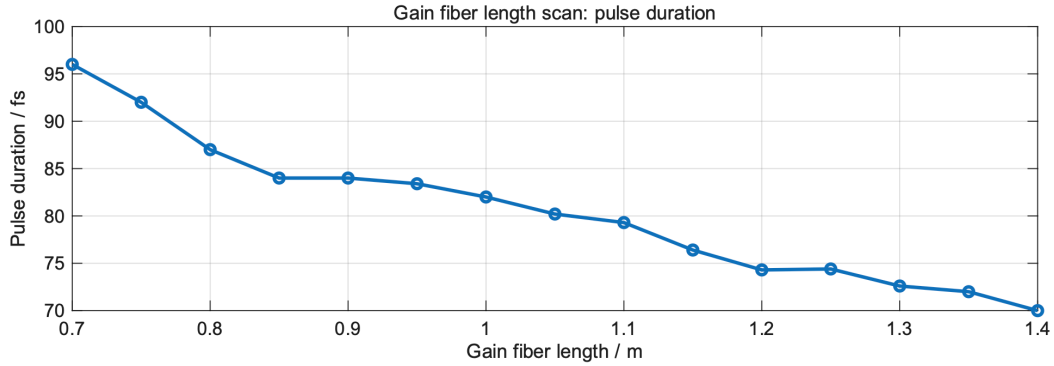



Figure 4-21 Pulse duration versus gain fiber length.

# 5 Conclusion

This paper conducts a study on the factors influencing the performance of Mamyshev oscillators through theoretical modeling and numerical simulation. Based on the generalized nonlinear Schrödinger equation, we build a simulation model for ring-cavity and linear-cavity Mamyshev oscillators. By varying the pump power, filter center wavelength spacing, filter bandwidth, and ytterbium-doped gain fiber length, the evolution of output pulse energy, peak power, pulse width, and center wavelength under different structures is analyzed. Furthermore, the differences in parameter response and stability between the two cavity types are compared.

For the ring-cavity Mamyshev oscillator, simulation results show that the formation of a stable mode-locking state depends on the dynamic balance between gain, dispersion, nonlinear spectral broadening, and offset spectral filtering. In the pump power scan, as the pump power increases from 3 W to 7 W, the output pulse energy and peak power generally trend upward, while the pulse width gradually decreases from approximately 61 fs to 38 fs. This indicates that higher pump power enhances gain accumulation and self-phase modulation (SPM) effects, allowing the pulse to achieve stronger spectral broadening and shorter output pulses under filtering. However, when the pump power continues to rise, the system struggles to maintain a stable single-pulse mode-locking state, indicating that excessive gain disrupts the balance, causing the oscillator to enter a multi-pulse or unstable regime. Therefore a balance between output energy and mode-locking stability is required.

The filter center wavelength spacing also significantly impacts the pulse-shaping process. As the spacing between the two filters increases, the system's requirement for SPM-induced spectral broadening increases; only high-intensity pulses with sufficient spectral broadening can pass through the offset filter, thus enhancing the equivalent saturable absorption effect. In the ring-cavity structure, as the filter spacing increases from 4 nm to 12 nm, the pulse width generally decreases, peak power gradually increases, and the output center wavelength undergoes a significant blue shift. Meanwhile, larger filter spacing also increases loss, causing pulse energy to decrease within a certain range. This shows that larger filter spacing is beneficial for obtaining narrower pulses and higher peak power, but excessive spacing may lead to overly strong filtering loss and reduced mode-locking stability.

The effect of filter bandwidth shows an obvious optimal range. In the ring-cavity simulation, when the filter bandwidth increases from 1 nm to 3 nm, more nonlinearly broadened spectral components can pass through the filter, increasing output pulse energy and peak power.

When the bandwidth continues to increase, the spectral selection effect of the filter weakens, the equivalent saturable absorption effect decreases, and the pulse-shaping capability deteriorates, causing the output energy and peak power to decrease instead. The pulse width also shows a trend of first decreasing and then increasing, indicating that a filter that is too narrow causes excessive spectral loss, while one that is too wide cannot provide sufficiently effective nonlinear screening. Therefore, in Mamyshev oscillator design, the filter bandwidth needs to balance energy throughput and pulse-shaping intensity.

Changes in gain fiber length mainly affect the gain and nonlinear phase shift acquired by the pulse in the cavity. In the ring-cavity structure, as the length of the ytterbium-doped gain fiber increases, pulse energy and peak power generally rise, and the pulse width gradually decreases. This is because longer gain fiber provides more sufficient pump absorption and gain accumulation while enhancing the spectral broadening caused by self-phase modulation, allowing the pulse to be more significantly compressed during the subsequent filtering process. However, increasing the gain fiber length also brings more dispersion and nonlinear phase accumulation, so it needs to be adjusted alongside the passive fiber length to maintain the balance between intracavity dispersion, nonlinearity, and filtering.

For the linear-cavity Mamyshev oscillator, simulation results show that under selected parameters, the linear cavity can produce de-chirped pulses of about 80 fs with a peak power of approximately 400 kW, demonstrating that this structure can also achieve stable ultrashort pulse output.

Regarding parameter trends, the linear cavity shares similar physical laws with the ring cavity. When pump power increases, pulse energy and peak power in the linear cavity also rise, and the pulse width generally decreases. When filter offset increases, the pulses undergo stronger nonlinear spectral filtering, resulting in shorter pulse widths and corresponding shifts in center wavelength. When the gain fiber length increases, the output energy increases and pulse width shortens. This indicates that whether a ring-cavity or linear-cavity structure is used, the fundamental mechanism of the Mamyshev oscillator stems from the coupling between self-phase modulation spectral broadening and offset filtering.

However, there are significant differences between the two cavity types in terms of parameter sensitivity. Pulses in a ring cavity typically have longer propagation paths and stronger nonlinear evolution processes, making it easier to achieve higher output energy and peak power. In contrast, the linear-cavity structure is more compact and suitable for all-fiber designs using FBG reflectors, but its stable operating state is more sensitive to FBG bandwidth and center wavelength spacing. In the linear cavity filter bandwidth scan, as the LR-FBG bandwidth in-

creases, pulse energy and peak power first rise and then fall, and the pulse width also shows a trend of first decreasing and then increasing—a change more pronounced than in the ring cavity. This suggests that the stable mode-locking window of the linear cavity is relatively narrow, and slight deviations in FBG parameters could weaken the filtering selection or disrupt intracavity balance. Thus, the linear-cavity Mamyshev oscillator requires more precise optimization of FBG parameters and fiber lengths in engineering design.

In summary, the simulation results of this paper indicate that the output performance of Mamyshev oscillators is not determined by a single parameter but by the coupling of gain, dispersion, nonlinearity, and spectral filtering. The ring-cavity structure is better suited for studying the generation mechanisms of high-energy, high-peak-power ultrashort pulses, while the linear-cavity structure has greater potential for all-fiber applications and engineering integration. The results obtained in this paper provide a theoretical reference for the structural design and parameter optimization of Mamyshev oscillators and offer a simulation basis for achieving more stable, higher-energy, and more easily integrated ultrafast fiber lasers.

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# 致　　谢

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